

The missing third-to-fourth cumulant feed

Gaussian spectral response, scale Ward identities, and observable closure

A constructive investigation of the Connes NCG and index-theory route

Result. The omitted fourth-cumulant response has an exact scalar influence formula and an exact Gaussian gate-facet representation. A Gaussian Bott–Dirac spectral triple places the response in the third Hermite sector; its ordinary index cancels that sector. Positive compression defects and reachable–observable ranks characterize the additional information needed for dynamic closure. A separate conditional factor theorem makes evaluation economical when the Gaussian reference has a small common-factor representation. **Status.** General analytic results with proofs. Exact

algebra checks and independent proof review are included. No network weights, raw ledger arrays, or repository were available for this investigation. The numerical deep-layer rate in networks 0 and 1 is not claimed to have been derived.

7 October 2026

Checkpoint NCG-20261007-A

Based on the user-supplied Note XIX and accompanying research notes.

Contents

1	The target and the results	1
1.1	What is established	1
1.2	How index theory enters, and the precise limit of the result	2
1.3	Reading guide and notation	2
2	An exact source-response theory for the omitted fourth-cumulant sectors	2
2.1	The contracted missing observable and all of its recentering terms	3
2.2	A specified Gaussian third-cumulant tangent	4
2.3	Bulk terms and gate facets, with unambiguous traces	5
2.4	The exact formula for the retained incoming pair slices	6
2.5	Affine cancellation and a quantitative gate-saturation bound	7
2.6	A full-rank analytic witness using a retained incoming pair slice	9
2.7	From the exact response to a gain entry	10
2.8	Independent exact algebra checks and their limits	10
3	An index-theoretic route, and the spectral information that survives it	11
3.1	The Gaussian differential complex	11
3.2	An actual algebra, and the regularity obstruction at a gate	13
3.3	The index pairing and what it records	14
3.4	A genuine source tangent and its third-chaos spectral support	15
3.5	Heat quantities retain information that the index cancels	16
3.6	A precise limitation of index-only formulas	17
3.7	A spectral gate diagnostic with exact regularity thresholds	19
3.8	The quantitative object for a contracted source ledger	20
3.9	Checked primary sources and the source-to-target dictionary	21
4	The exact radial Ward identity	22
5	Compression calculus and moving retained frames	24
6	An exact operator criterion for retained-observable closure	26
6.1	Reference laws, tangent scores, and the direction of the operators	26
6.2	Closure, coefficient error, and intrinsic projection leakage	26
6.3	A positive Gram defect and the number of missing observables	27
6.4	Why a finite projection index cannot replace the defect	28
7	Exact finite-depth memory and minimal observable state	29
7.1	Block elimination without a stationarity assumption	29
7.2	Schur complement and the self-energy of an omitted channel	30

7.3	Minimal reachable–observable realization over a finite horizon	31
7.4	The exact finite-horizon query space	32
8	Readout-weighted error and a finite test of insufficiency	32
8.1	An exact adjoint error identity	32
8.2	A finite probe criterion with an explicit scope	33
8.3	What remains specific to the reported networks	33
9	An exact factor-Gaussian reduction of the missing pair-source response	34
9.1	The scalar observable and its complete frozen influence	34
9.2	Univariate half moments and their derivative jets	35
9.3	Seven conditional coordinates suffice for the scalar polynomial	35
9.4	The incoming pair-source coefficients	36
9.5	Baseline constants require only scalar, vector, and pair contractions	37
9.6	What this constructs, and what remains a separate theorem	38
10	The duality bridges and the remaining network theorem	40
10.1	A precise bridge from observable logic to dynamic closure	40
10.2	The noncommutative object that measures the lost response	40
10.3	What the next theorem for these networks must establish	41
10.4	Evidence, review, and scope of the closeout	42

1 The target and the results

The supplied Note XIX reports that the pair-slice third-cumulant transport is already accurate, while the analogous fourth-cumulant contraction loses a substantial coherent component at depth. Its retained fourth sectors are the index-multiplicity classes (4) and (2, 2). The missing classes are (3, 1), (2, 1, 1), and (1, 1, 1, 1). The present target is the response of their *weighted sum* to the same incoming unit third-gain tangent used in the ledger. The target is not an arbitrary completion of a fourth-order tensor.

For a reference preactivation vector Z , one ReLU layer $H_i = (Z_i)_+$, a fixed next weight row w , and an incoming source direction T , write

$$\Delta_T(w) = \delta_T K_{\text{miss}}(w).$$

After calibrating T to one incoming gain unit and applying the specified outgoing coherent projection, these row responses give the missing contribution to r_{43} . The centering, covariance, gain denominator, and any fitted projection are part of that derivative. If those coordinates move, their derivative terms must be included.

The supplied notes are evidence of the project's reported observations. They are not the raw network arrays, and this note does not independently revalidate their measured percentages. The statements below separate the following three mathematical levels:

1. **One-step source response:** a derivative at a specified reference law and specified incoming tangent.
2. **Dynamic closure:** whether retained observables determine all required future derivatives for a declared class of perturbations.
3. **Economical evaluation:** whether the exact formulas can be computed within the intended width, depth, and accuracy budget.

An exact formula at the first level does not imply the other two.

1.1 What is established

Result	Mathematical content and practical consequence
Missing response	A scalar polynomial influence includes all three omitted sectors and every centering/covariance derivative. Its Gaussian third-source response is $\mathbb{E}[h_T \Psi_{\text{miss}}]$. No outgoing fourth-order array is required.
Gate geometry	The third weak derivative consists of bulk terms, single gate facets, and normal derivatives of facet measures. At this order no double-boundary terms appear. A matched incoming pair-source formula is explicit.
Actual NCG construction	The Gaussian Bott–Dirac operator has compact resolvent and index one. The desired response is a third-chaos matrix element, equivalently a pole residue of a weighted resolvent. That residue is not a local index residue.
Scale Ward identity	Positive homogeneity gives an exact law-level identity. At a Gaussian input, the regular scale tangent has only first through fourth cumulant components. Reference drift and output normalization complete the two-channel relation.

Result	Mathematical content and practical consequence
Closure obstruction	The positive defect of compressed propagation vanishes exactly when the retained tangent observables close. Its singular values give optimal local augmentation errors; a finite-horizon reachable–observable quotient gives exact minimal linear memory.
Conditional evaluation	Under an explicit diagonal-plus- r -factor Gaussian representation, the matched pair-source response is an r -dimensional integral whose integrand costs $O(n^2)$ per row. Small r is an additional hypothesis, not a consequence of two observed gain channels.

1.2 How index theory enters, and the precise limit of the result

Connes’ index pairing associates an integer to an appropriate Fredholm cycle and a K -theory class. The entire Chern character gives a heat-operator expression for that pairing under verified summability hypotheses [1, 2]. We construct those data explicitly. This is stronger than using the word “noncommutative” for an order-dependent approximation.

The construction then shows why an ordinary index alone cannot determine the missing continuous amplitude. All positive Gaussian energies cancel in the unweighted supertrace, including the third-chaos sector containing the incoming source. Weighted spectral matrix elements retain the missing information. The result leaves open other constructions using measured indices, varying traces, or relative classes; such a route would need an explicit comparison theorem with this particular cumulant response.

The local index formula of Connes–Moscovici requires additional analytic regularity and spectral hypotheses [3]. We do not assume them for gate indicators. In fact, the exact gate projection fails the first Dirac-domain condition in this construction. ReLU and its gate therefore require different treatment.

1.3 Reading guide and notation

Section 2 derives the missing response. Section 3 builds the spectral and index theory. Section 4 proves the scale identity. Section 5 treats compression and changing coordinate frames. Section 6 proves closure and memory statements. Section 9 gives the conditional evaluation theorem. Section 10 reconnects these results to the duality programme and states the remaining network-specific obligation.

Symbols are local to their sections. In particular, D denotes a scalar covariance trace in the response section, a Dirac operator in the spectral section, and an omitted-state propagation block in the memory section. The influence polynomial called F in the response section is denoted Ψ_{miss} in this introduction. It is not a layer map. All multiplicity-sector sums are ordered unless stated otherwise.

2 An exact source-response theory for the omitted fourth-cumulant sectors

This section gives an exact one-step answer to a specified question. Fix the Gaussian reference state, the incoming third-cumulant tangent, the next weight row, and the gain projection. What is the first-order contribution of the omitted output sectors $(3, 1)$, $(2, 1, 1)$, and $(1, 1, 1, 1)$? The answer below requires neither an outgoing fourth-order tensor nor a fitted coefficient. It does not assert that a Gaussian tangent is a complete model of the actual non-Gaussian input law, or that two gain coordinates form an autonomous state.

The incoming direction must be the one used by the existing ledger. Restoring outgoing sectors while simultaneously adding previously omitted incoming third-cumulant components would answer a different question.

2.1 The contracted missing observable and all of its recentering terms

Let $H = (H_1, \dots, H_n)$ have finite fourth moments, let $w \in \mathbb{R}^n$ be fixed, and put

$$u_i = w_i(H_i - \mathbb{E}H_i), \quad S_r = \sum_i u_i^r, \quad C_{ij} = \mathbb{E}[u_i u_j], \quad (1)$$

$$V = \mathbf{1}^\top C \mathbf{1} = \text{Var}(S_1), \quad D = \text{tr } C, \quad R = \sum_{i \neq j} C_{ij}^2. \quad (2)$$

Here and below the sum defining R is ordered. Write C_{off} for C with its diagonal set to zero. The retained contribution in Note XIX is the sum of the (4) and (2, 2) index-multiplicity classes. Consequently the complement is exactly

$$K_{\text{miss}} = \sum_{(i,j,k,l) \in \mathcal{I}_{\text{miss}}} \text{cum}(u_i, u_j, u_k, u_l), \quad \mathcal{I}_{\text{miss}} = (31) \cup (211) \cup (1111). \quad (3)$$

This definition fixes the neuron coordinate system. Individual multiplicity classes are not invariant under arbitrary changes of hidden basis.

Proposition 1 (contracted identity). Set

$$P(u) = S_1^4 - 3S_2^2 + 2S_4. \quad (4)$$

Then

$$\boxed{K_{\text{miss}} = \mathbb{E}P(u) - 3(V^2 - D^2) + 6R.} \quad (5)$$

In particular, this is an identity of scalar contractions, not a construction of a fourth-order tensor.

Proof. The complete fourth moment is S_1^4 . Its (4) and (2, 2) contributions sum to $S_4 + 3(S_2^2 - S_4) = 3S_2^2 - 2S_4$. For these retained classes the Gaussian covariance-pairing terms sum to $3D^2 + 6R$. The complete pairing term is $3V^2$. Subtract the retained cumulant from $\mathbb{E}S_1^4 - 3V^2$. $\square$

Let $\mathcal{L}$ be a first variation of the source law with $\mathcal{L}\mathbf{1} = 0$. We assume differentiability of the displayed moments and keep the readout w fixed. Define the baseline constants

$$b_i = \mathbb{E}[\partial_i P(u)] = 4\mathbb{E}S_1^3 - 12\mathbb{E}[S_2 u_i] + 8\mathbb{E}u_i^3. \quad (6)$$

Theorem 2 (exact missing-sector influence polynomial). With all constants on the right evaluated at the reference law, put

$$\boxed{F(u) = P(u) - b^\top u - 6VS_1^2 + 6DS_2 + 12u^\top C_{\text{off}}u.} \quad (7)$$

Then the complete first variation is

$$\boxed{\delta K_{\text{miss}} = \mathcal{L}[F(u)].} \quad (8)$$

An additive constant in F has no effect. No term arising from the changing mean or covariance has been frozen out of this identity.

Proof. Write $\beta_i = \mathcal{L}u_i$. As the source law changes, the centered coordinate has first variation $\delta u_i = -\beta_i$. Thus

$$\delta \mathbb{E}P(u) = \mathcal{L}P(u) - \sum_i \beta_i \mathbb{E} \partial_i P(u) = \mathcal{L}[P(u) - b^\top u].$$

Since $\mathbb{E}u_i = 0$, the derivative of C_{ij} is simply $\delta C_{ij} = \mathcal{L}(u_i u_j)$. It follows that

$$\delta V = \mathcal{L}S_1^2, \quad \delta D = \mathcal{L}S_2, \quad \delta R = 2 \sum_{i \neq j} C_{ij} \mathcal{L}(u_i u_j).$$

Differentiate (5) and collect these linear functionals. $\square$

The coefficients of F involve one scalar third moment, the contractions $\mathbb{E}(S_2 u_i)$ and $\mathbb{E}u_i^3$, and the weighted covariance. They do not require an outgoing third- or fourth-order tensor. Computing an exact scalar contraction can nevertheless be expensive; this theorem alone gives no bound on the cost of an exact Gaussian orthant integral.

For use below, write $F_i = \partial_i F$, and similarly for higher derivatives. Direct differentiation gives

$$F_i = 4S_1^3 - 12S_2 u_i + 8u_i^3 - b_i - 12V S_1 + 12D u_i + 24(C_{\text{off}} u)_i, \quad (9)$$

$$F_{ii} = 12(S_1^2 - S_2 - V + D), \quad (10)$$

$$F_{ij} = 12S_1^2 - 24u_i u_j - 12V + 24C_{ij} \quad (i \neq j), \quad (11)$$

$$F_{ijk} = 24S_1 - 24(\delta_{ik} u_j + \delta_{jk} u_i + \delta_{ij} u_k) + 48\delta_{ij} \delta_{ik} u_i. \quad (12)$$

In particular,

$$F_{iii} = 24(S_1 - u_i), \quad F_{iij} = 24(S_1 - u_j) \quad (i \neq j). \quad (13)$$

2.2 A specified Gaussian third-cumulant tangent

Let $Z \sim N(\mu, \Sigma)$ with Σ positive definite, let $H_i = \rho(Z_i)$ for $\rho(t) = \max(t, 0)$, and use the preceding definitions of u and F . Put

$$Q = \Sigma^{-1}, \quad q = Q(Z - \mu). \quad (14)$$

For a symmetric tensor T specifying the incoming third-cumulant first variation, define

$$h_T(Z) = \frac{1}{6} \sum_{ijk} T_{ijk} (q_i q_j q_k - Q_{ij} q_k - Q_{ik} q_j - Q_{jk} q_i). \quad (15)$$

The pure third-order tangent is $\mathcal{L}_T G = \mathbb{E}[h_T G]$. Gaussian integration by parts yields, in the distributional sense,

$$\mathcal{L}_T G = \frac{1}{6} \sum_{ijk} T_{ijk} \mathbb{E}[\partial_{ijk} G]. \quad (16)$$

The distributional qualification is essential for ReLU.

Proposition 3 (exact source-response formula). The missing fourth-order response to this specified tangent is

$$\Delta_T = \delta_T K_{\text{miss}} = \mathbb{E}[h_T(Z) F(u(Z))]. \quad (17)$$

Moreover, the tangent changes the source third cumulant by T and changes no other source cumulant at first order.

Proof. The first assertion is Theorem 2. For the second, Gaussian integration by parts applied to $e^{t^\top Z}$ gives

$$\frac{\mathbb{E}[h_T e^{t^\top Z}]}{\mathbb{E}e^{t^\top Z}} = \frac{1}{6} T[t, t, t].$$

This is the first variation of the cumulant-generating function. $\square$

This is a genuine infinitesimal probability direction. For example,

$$p_\epsilon(z) = p(z) \frac{(1 + \epsilon h_T(z)/2)^2}{1 + \epsilon^2 \mathbb{E} h_T^2/4} \quad (18)$$

is nonnegative and normalized, and its derivative at zero is ph_T . The linear expression $p(1 + \epsilon h_T)$ need not itself be a probability density for nonzero ϵ .

Prescribing only a third-cumulant change does not uniquely prescribe a law perturbation: one may add higher Hermite components while preserving that third-cumulant change. Equation (17) is exact for the specified pure third-order tangent, not for every law with the same incoming third cumulant. A singular Gaussian must first be represented on its affine support; an arbitrary T cannot be handled merely by replacing an inverse by a pseudoinverse.

2.3 Bulk terms and gate facets, with unambiguous traces

We next give a direct geometric form of the response. It makes the gate terms explicit and avoids products such as $\mathbf{1}_{z_i > 0} \delta(z_i)$, whose informal manipulation can lose coefficients.

Fix a direction $d \in \mathbb{R}^n$ and let

$$G_i(z) = \mathbf{1}_{z_i > 0}, \quad a_i(z) = w_i d_i G_i(z), \quad \partial_d = \sum_i d_i \partial_{z_i}. \quad (19)$$

For any expression $A(z)$ continuous in z_i at zero, $A^{(i)}$ denotes its trace obtained by setting $z_i = 0$. A coefficient multiplying δ'_i is extended constantly in the normal coordinate, so that it depends only on z_{-i} . Let $\delta_i = \delta(z_i)$ and $\delta'_i = \delta'(z_i)$.

Theorem 4 (third derivative on one ReLU layer). For a C^3 function F whose derivatives through order three have polynomial growth in u , the following identity of distributions holds:

$$\begin{aligned} \partial_d^3[F(u(z))] &= \sum_{ijk} F_{ijk}(u) a_i a_j a_k \\ &+ \sum_i \left[w_i^2 d_i^3 F_{ii} + 3w_i d_i^2 \sum_{j \neq i} a_j F_{ij} \right]^{(i)} \delta_i \\ &+ \sum_i w_i d_i^3 F_i^{(i)} \delta'_i. \end{aligned} \quad (20)$$

At this derivative order there are bulk terms and codimension-one facet terms. There are no products $\delta_i \delta_j$ for $i \neq j$.

Proof. The function $F(u(z))$ is continuous across every coordinate hyperplane. Its first weak derivative is therefore the ordinary piecewise derivative $\sum_i F_i a_i$. Across the i th hyperplane the jump of this first derivative is $w_i d_i F_i^{(i)}$. Hence

$$\partial_d^2 F(u) = \sum_{ij} F_{ij} a_i a_j + \sum_i w_i d_i^2 F_i^{(i)} \delta_i.$$

The jump of the bulk term in this expression across $z_i = 0$ is

$$w_i^2 d_i^2 F_{ii}^{(i)} + 2w_i d_i \sum_{j \neq i} a_j F_{ij}^{(i)}.$$

Its normal derivative gives the first two contributions to the δ_i coefficient in (20). The tangential derivative of $F_i^{(i)}$ contributes one more copy of $\sum_{j \neq i} a_j F_{ij}^{(i)}$, giving the factor 3. The normal

derivative of δ_i gives $d_i \delta'_i$. Finally, $F_i^{(i)}$ is continuous across the other coordinate hyperplanes, so its first tangential derivative produces no additional delta distribution. This proves both the formula and the absence of codimension-two terms at order three. $\square$

The theorem concerns one coordinatewise ReLU layer followed by the chosen polynomial observable. It is not an assertion about all strata of an arbitrary multi-layer piecewise-linear map.

Let $p_i(0)$ be the marginal density of Z_i at zero and let $\mathbb{E}_i$ denote conditional expectation under $Z_i = 0$. The Gaussian density satisfies $\partial_i p = -q_i p$, and consequently

$$\mathbb{E}[A^{(i)} \delta_i] = p_i(0) \mathbb{E}_i A^{(i)}, \quad \mathbb{E}[A^{(i)} \delta'_i] = p_i(0) \mathbb{E}_i [q_i A^{(i)}]. \quad (21)$$

The positive sign in the second identity comes from integration by parts. Combining Theorem 4 with (16) gives a scalar bulk expectation and conditional Gaussian facet expectations. It requires no outgoing cumulant tensor.

For illustration, the total scalar fourth cumulant of $Y = w^\top H$ has the influence polynomial $\psi(S) = S^4 - 6VS^2 - 4c_3S$, where $S = Y - \mathbb{E}Y$ and $c_3 = \mathbb{E}S^3$. Writing $A = \sum_i a_i$ and $A_{-i} = \sum_{j \neq i} a_j$, the formula becomes

$$\begin{aligned} 6\delta_{d \otimes 3} \kappa_4(Y) &= 24\mathbb{E}[SA^3] \\ &+ \sum_i p_i(0) \mathbb{E}_i \left[12w_i d_i^2 (w_i d_i + 3A_{-i})(S^2 - V) + w_i d_i^3 q_i (4S^3 - 12VS - 4c_3) \right]. \end{aligned} \quad (22)$$

The bulk term is generally nonzero. Gate-boundary language must not be used to discard it.

2.4 The exact formula for the retained incoming pair slices

The existing source direction can be specified by a diagonal vector and an oriented pair table,

$$\tau_i = T_{iii}, \quad \Gamma_{ij} = T_{ijj} \quad (i \neq j), \quad \Gamma_{ii} = 0, \quad (23)$$

with all fully distinct incoming entries zero. The full symmetry of T includes the three permutations of (i, i, j) ; it does not imply $\Gamma_{ij} = \Gamma_{ji}$.

Corollary 5 (matched pair-source formula). For precisely the source in (23), the omitted output response is

$$\begin{aligned} \Delta_T &= \frac{1}{6} \sum_i \tau_i \left\{ w_i^3 \mathbb{E}[G_i F_{iii}] + p_i(0) \mathbb{E}_i [w_i^2 F_{ii} + w_i q_i F_i] \right\} \\ &+ \frac{1}{2} \sum_{i \neq j} \Gamma_{ij} \left\{ w_i^2 w_j \mathbb{E}[G_i G_j F_{ijj}] + p_i(0) w_i w_j \mathbb{E}_i [G_j F_{ij}] \right\}. \end{aligned} \quad (24)$$

Here F is exactly (7), and the explicit derivatives are (9)–(13). All facet expressions use their $z_i = 0$ traces.

Proof. For a repeated derivative in one coordinate,

$$\partial_i^3 F(u) = w_i^3 G_i F_{iii} + w_i^2 F_{ii}^{(i)} \delta_i + w_i F_i^{(i)} \delta'_i.$$

For $i \neq j$,

$$\partial_i^2 \partial_j F(u) = w_i^2 w_j G_i G_j F_{ijj} + w_i w_j G_j F_{ij}^{(i)} \delta_i.$$

Insert these formulas into the Gaussian third-order derivative. Each T_{ijj} appears three times, giving $3/6 = 1/2$. Finally apply (21). $\square$

The same response can be evaluated by the score

$$h_T = \frac{1}{6} \sum_i \tau_i (q_i^3 - 3Q_{ii}q_i) \quad (25)$$

$$+ \frac{1}{2} \sum_{i \neq j} \Gamma_{ij} (q_i^2 q_j - Q_{ii} q_j - 2Q_{ij} q_i). \quad (26)$$

Thus even the incoming direction need not be materialized as a cubic array. For example, the double sum is

$$\frac{1}{2} \{ (q^{\odot 2} - \text{diag } Q)^\top \Gamma q - 2q^\top [(\Gamma \odot Q) \mathbf{1}] \}.$$

This is an algebraic cost reduction, not a guarantee that a dense precision solve and pair-table multiplication are inexpensive at competition scale.

2.5 Affine cancellation and a quantitative gate-saturation bound

Proposition 6 (affine limit). If each activation is replaced by a fixed affine map, the response of every outgoing fourth-cumulant component to the pure third-cumulant source tangent is zero. In particular, $\Delta_T = 0$.

Proof. An affine map substitutes a linear argument into the source cumulant-generating function. Its first variation, by Proposition 3, is a homogeneous cubic polynomial. Its fourth derivatives are therefore zero. $\square$

For the actual ReLU map, we now give an explicit bound showing how the response tends to zero when every marginal gate becomes saturated. The bound is deliberately conservative, but its constants and limitations are visible. Let

$$\sigma_i^2 = \Sigma_{ii}, \quad r_i = |\mu_i|/\sigma_i, \quad c_p = (\mathbb{E}|N(0, 1)|^p)^{1/p}, \quad (27)$$

$$B = \sum_i |w_i| \sigma_i, \quad L = \sum_i |w_i d_i|, \quad A_p(r) = r + c_p + c_1, \quad (28)$$

$$C_3 = (c_3 + c_1)^3, \quad \chi_i = \left(\frac{r_i^2}{\sigma_i^2} + Q_{ii} - \frac{1}{\sigma_i^2} \right)^{1/2}. \quad (29)$$

In this subsection c_3 denotes the absolute standard-normal moment norm, not an output cumulant. Write φ for the standard-normal density.

Theorem 7 (explicit density bound). For the rank-one source $T = d^{\otimes 3}$,

$$\begin{aligned} |\Delta_T| \leq \frac{1}{6} & \left[1296BL^2 \sum_i |w_i d_i| \varphi(r_i) \right. \\ & + \sum_i \frac{\varphi(r_i)}{\sigma_i} \left\{ B^2 |w_i| |d_i|^2 (|w_i d_i| + 3L) [72A_2(r_i)^2 + 48] \right. \\ & \quad \left. \left. + B^3 |w_i| |d_i|^3 \chi_i [24A_6(r_i)^3 + 48A_2(r_i) + 24C_3] \right\} \right]. \end{aligned} \quad (30)$$

For $T = \sum_s \lambda_s d_s^{\otimes 3}$, sum the corresponding bounds with weights $|\lambda_s|$. No assertion of a small rank is needed for the theorem, and no small-rank representation follows from it.

In particular, for fixed positive-definite Σ , fixed w , and a fixed finite source decomposition, the unnormalized missing response tends to zero as $\min_i r_i \rightarrow \infty$. Every term on the right is a Gaussian boundary density times a polynomial in the corresponding gate margin.

Proof. Let $M = \sum_i |u_i|$. The one-Lipschitz property of ReLU gives

$$|H_i - \mathbb{E}H_i| \leq \sigma_i(|\xi_i| + c_1), \quad \xi_i = (Z_i - \mu_i)/\sigma_i.$$

Therefore $\|M\|_p \leq B(c_p + c_1)$. Conditionally on $Z_i = 0$, each standardized coordinate has mean of absolute value at most r_i and variance at most one; hence $\|M\|_{p,i} \leq BA_p(r_i)$.

The elementary independent-copy identity for variance and the Lipschitz property imply $\text{Var}(H_i) \leq \sigma_i^2$. Consequently $V \leq B^2$, $D \leq B^2$, and $|C_{ij}| \leq |w_i|\sigma_i|w_j|\sigma_j$. The explicit derivatives of F yield the pointwise bounds

$$\begin{aligned} |F_{ij}| &\leq 72M^2 + 48B^2, \\ |F_i| &\leq 24M^3 + 48B^2M + 24B^3C_3. \end{aligned}$$

In the second line, $|b_i| \leq 24\mathbb{E}M^3$ was used. Also

$$\|q_i\|_{2,i} = \chi_i,$$

because the conditional mean is $-\mu_i/\sigma_i^2$ and the conditional variance is $Q_{ii} - \sigma_i^{-2}$. Cauchy–Schwarz, the displayed conditional moment bounds, and (20) prove exactly the two facet terms in (30).

For completeness, the bulk term needs a separate argument; it is not a facet distribution. Set $\theta_i = w_id_i$, $a_i = \theta_iG_i$, and choose fixed gates $G_i^* = \mathbf{1}_{\mu_i > 0}$ (either choice at $\mu_i = 0$ is harmless). Let $a_i^* = \theta_iG_i^*$. Contracting (12) gives

$$\begin{aligned} \sum_{ijk} F_{ijk} a_i a_j a_k &= 24S_1 \left(\sum_i a_i \right)^3 - 72 \left(\sum_i a_i^2 \right) \left(\sum_i u_i a_i \right) + 48 \sum_i u_i a_i^3 \\ &= \sum_j u_j f_j(a), \end{aligned}$$

where $f_j(a) = 24(\sum_i a_i)^3 - 72(\sum_i a_i^2)a_j + 48a_j^3$. On the ℓ^1 ball of radius L , every coordinate partial derivative of f_j has absolute value at most $432L^2$. Since $\mathbb{E}u_j = 0$,

$$\left| \mathbb{E} \sum_j u_j f_j(a) \right| \leq 432L^2 \sum_i |\theta_i| \mathbb{E}[M \mathbf{1}_{G_i \neq G_i^*}].$$

Conditional Gaussian regression and the preceding Lipschitz bound give

$$\mathbb{E}[M \mathbf{1}_{G_i \neq G_i^*}] \leq B\{\varphi(r_i) + 2c_1\bar{\Phi}(r_i)\} \leq 3B\varphi(r_i).$$

The last step uses $\bar{\Phi}(r) \leq \sqrt{\pi/2} \varphi(r)$ for $r \geq 0$, which follows by writing the ratio as $\int_0^\infty e^{-rs-s^2/2} ds$. This establishes the coefficient 1296 in the bulk bound. Finally the Gaussian derivative formula contributes the factor $1/6$. $\square$

The bound is for the raw fourth-cumulant response. A normalized response dividing by the output variance squared need not vanish if that variance also tends to zero, as can occur when every gate turns off. A uniform conclusion about normalized gains requires a positive lower bound on the normalization denominator and control of its derivative. The constants also expose dependence on the precision matrix; no uniform bound over arbitrarily ill-conditioned Gaussian references has been asserted.

2.6 A full-rank analytic witness using a retained incoming pair slice

This example isolates omitted outgoing information while keeping the incoming source within the existing pair-slice class.

Let Z_1, Z_2 be independent standard Gaussians, $H_i = \rho(Z_i)$, and let

$$T_{112} = T_{121} = T_{211} = t, \quad h_T = \frac{t}{2}(Z_1^2 - 1)Z_2,$$

with all other source entries zero. Put $c = \varphi(0) = 1/\sqrt{2\pi}$. At the reference law H_1, H_2 are independent, so every mixed fourth cumulant is zero. Nevertheless their first variations need not vanish.

Proposition 8 (nonzero omitted response). For this source,

$$\delta \text{cum}(H_1, H_1, H_1, H_2) = t \left(\frac{3c}{8} + \frac{3c^3}{2} \right), \quad (31)$$

$$\delta \text{cum}(H_1, H_2, H_2, H_2) = t \left(\frac{3c}{8} - \frac{3c^3}{2} \right). \quad (32)$$

For $w = (1, 1)$, only the $(3, 1)$ omitted class exists and

$$\boxed{\Delta_T = 3tc.} \quad (33)$$

The retained $(2, 2)$ response is $3tc(1 - 2c^2)$, and the complete scalar fourth cumulant response is $6tc(1 - c^2)$.

Proof. For $U = \rho(Z) - c$,

$$v = \mathbb{E}U^2 = \frac{1}{2} - c^2, \quad k_3 = \mathbb{E}U^3 = \frac{c}{2} + 2c^3.$$

At the independent reference, differentiating the joint fourth cumulant gives

$$\delta \text{cum}(H_1, H_1, H_1, H_2) = \frac{t}{2} \mathbb{E}[(Z^2 - 1)(U^3 - 3vU - k_3)] \mathbb{E}[ZU].$$

The half-normal moments give

$$\mathbb{E}[ZU] = \frac{1}{2}, \quad \mathbb{E}[(Z^2 - 1)U] = c, \quad \mathbb{E}[(Z^2 - 1)U^3] = 3c + 3c^3.$$

This proves the first line of (32). For the second, exchange which factor carries the cubic output polynomial and use

$$\mathbb{E}[ZU^3] = \frac{3}{2} - \frac{9}{2}c^2.$$

There are four ordered tuples of each $(3, 1)$ type, so their sum is $3tc$. For the $(2, 2)$ component the marginal variance derivatives vanish and the baseline cross-covariance is zero. Its derivative is

$$\frac{t}{2} \mathbb{E}[(Z^2 - 1)U^2] \mathbb{E}[ZU^2] = \frac{tc}{2}(1 - 2c^2).$$

There are six ordered tuples, giving the stated retained response. The diagonal fourth-cumulant responses are zero, which proves the total. $\square$

For a general row $w = (a, b)$ the omitted response is

$$\boxed{\Delta_T(a, b) = \frac{3tc}{2} ab \{a^2(1 + 4c^2) + b^2(1 - 4c^2)\}.} \quad (34)$$

It varies continuously and nontrivially with the readout weights, with the same Gaussian source, the same gate partition, and the same incoming pair direction. This supplies an analytic obstruction to obtaining its numerical value from discrete gate-incidence information alone.

There is also a one-dimensional opposed-gate witness. Let $A = Z_+$, $B = (-Z)_+$, with $Z \sim N(0, 1)$, and use the source score $(Z^3 - 3Z)/6$. Direct half-normal moment calculation gives

$$\delta \text{cum}(A, A, A, B) = c(2c^2 - \tfrac{1}{2}), \quad \delta \text{cum}(A, B, B, B) = -c(2c^2 - \tfrac{1}{2}), \quad \delta \text{cum}(A, A, B, B) = 0.$$

Thus for $Y = aA + bB$,

$$\delta K_{\text{miss}} = 4ab(a^2 - b^2)c(1/\pi - \tfrac{1}{2}). \quad (35)$$

This second witness is computed on its one-dimensional Gaussian source, so it makes no illegitimate use of an inverse for the singular two-coordinate preactivation vector $(Z, -Z)$.

2.7 From the exact response to a gain entry

Suppose the missing fourth-cumulant vector across next-layer rows is k_{miss} , and the coherent gain is the fixed projection $a^\top k$. The missing contribution to the third-to-fourth transport entry is then

$$\Delta r_{43} = a^\top \delta_T k_{\text{miss}},$$

where T is calibrated to one unit of the existing third-gain coordinate. If the projection changes along the same source variation, the exact formula instead includes

$$\delta(a^\top k_{\text{miss}}) = a^\top \delta k_{\text{miss}} + (\delta a)^\top k_{\text{miss}}. \quad (36)$$

For the scalar convention $g_4 = \kappa_4/(3V^2)$, the corresponding quotient rule is

$$\delta g_4 = \frac{\delta \kappa_4}{3V^2} - 2g_4 \frac{\delta V}{V}.$$

Conditional-state rebasing, marginal-state evaluation, and normalization are part of this specified map. Agreement of their coordinate names does not make their derivatives equal.

The exact formulas provide a proof-level one-step observable and its source response. They do not establish an autonomous two-dimensional recursion: the response still depends on the chosen reference state and source direction. Nor do they prove that the omitted output response is the only deep error. Higher incoming cumulants, a wrong tangent model, state drift, and moving templates remain distinct terms. These distinctions prevent a correct one-step formula from being misreported as a complete derivation of the measured depth profile.

2.8 Independent exact algebra checks and their limits

The executable `checks/gate_response_exact.py` uses only Python standard-library rational arithmetic. It checks the closed bulk directional derivative as a polynomial identity in independent u_i, a_i for $n = 1, 2, 3, 4$. It also computes the two analytic witnesses independently from the set-partition definition of a joint cumulant, using exact Gaussian half-line moments in the ring $\mathbb{Q}[c]$ with $c = 1/\sqrt{2\pi}$. All six bounded checks passed.

The run is recorded in `computations/gate-response-exact/manifest.json`, with its declared input hash, command, output hash, resource limits, and Python version 3.12.14. The manifest validator accepted the evidence record, including input/output freshness. The recorded scientific output is `computations/gate-response-exact/results.json`. These checks concern exact algebra in the stated range. The proofs above, not a finite success flag, establish the distributional facet identity and the Gaussian saturation bound. No network weights, Monte Carlo stream, fitted gain table, or competition run were accessed.

3 An index-theoretic route, and the spectral information that survives it

This section constructs an actual even spectral triple on the Gaussian reference space, proves its index, and identifies precisely which analytic quantity contains the third-to-fourth-cumulant response. The construction is finite dimensional in its Gaussian input dimension but infinite dimensional as an operator representation. It is a standard Gaussian realization of the Bott–Dirac oscillator, with a direct proof included below. It supplies a legitimate application of Connes’ index pairing and entire cyclic cohomology. It does not identify a network transport eigenvalue with a topological index.

All Gaussian coordinates in this section are whitened. Thus a nondegenerate physical reference law $N(\mu, \Sigma)$ is first written as $Z_{\text{phys}} = \mu + \Sigma^{1/2}Z$, where $Z \sim \gamma_n = N(0, I_n)$. The network, output row, and source tensor must be transformed consistently. The index below depends on the reference differential complex, not on the particular neural weights. No numerical assertion about the supplied networks is made here.

3.1 The Gaussian differential complex

Put

$$d\gamma_n(x) = (2\pi)^{-n/2} e^{-|x|^2/2} dx, \quad \mathcal{H} = L^2(\gamma_n) \otimes \Lambda^* \mathbb{C}^n, \quad n \geq 1.$$

Let ε_j denote exterior multiplication by the j th basis vector and $\iota_j = \varepsilon_j^*$ interior multiplication. They satisfy

$$\varepsilon_i \varepsilon_j + \varepsilon_j \varepsilon_i = 0, \quad \iota_i \iota_j + \iota_j \iota_i = 0, \quad \iota_i \varepsilon_j + \varepsilon_j \iota_i = \delta_{ij} I.$$

On polynomial forms define

$$a_j = \partial_j, \quad a_j^* = x_j - \partial_j, \quad d_\gamma = \sum_j \varepsilon_j a_j, \quad d_\gamma^* = \sum_j \iota_j a_j^*, \quad D = d_\gamma + d_\gamma^*.$$

Gaussian integration by parts proves the stated adjoint relation for a_j . The bosonic and fermionic number operators are

$$N_b = \sum_j a_j^* a_j = -\Delta + x \cdot \nabla, \quad N_f = \sum_j \varepsilon_j \iota_j.$$

The grading is $\Gamma = (-1)^{N_f}$.

Proposition I.1 (the oscillator identity and its domain). The operator D on polynomial forms is essentially self-adjoint. Its closure satisfies

$$\boxed{D^2 = N_b + N_f, \quad \Gamma D = -D \Gamma.}$$

Its domain is the space of forms whose scalar coefficients belong to the Gaussian Sobolev space $W^{1,2}(\gamma_n)$, and

$$\|D\psi\|^2 = \sum_I \int |\nabla \psi_I|^2 d\gamma_n + \sum_I |I| \int |\psi_I|^2 d\gamma_n.$$

Proof. The canonical commutation relations $[a_i, a_j^*] = \delta_{ij} I$, together with the exterior-algebra relations, give $d_\gamma^2 = (d_\gamma^*)^2 = 0$ and

$$\begin{aligned} d_\gamma d_\gamma^* + d_\gamma^* d_\gamma &= \sum_{ij} \varepsilon_i \iota_j a_i a_j^* + \sum_{ij} \iota_j \varepsilon_i a_j^* a_i \\ &= \sum_i a_i^* a_i + \sum_i \varepsilon_i \iota_i. \end{aligned}$$

Let h_α be the normalized probabilists' Hermite polynomials. Then

$$a_j h_\alpha = \sqrt{\alpha_j} h_{\alpha - e_j}, \quad a_j^* h_\alpha = \sqrt{\alpha_j + 1} h_{\alpha + e_j}, \quad N_b h_\alpha = |\alpha| h_\alpha.$$

The Hermite polynomials form an orthonormal basis: orthogonality and the ladder identities follow by Gaussian integration by parts; completeness follows, for example, because a function orthogonal to every polynomial has identically zero Gaussian-weighted Laplace transform and hence zero Fourier transform. The Laplace transform in this argument exists and is analytic by Cauchy–Schwarz and Gaussian exponential integrability.

The finite dimensional spaces

$$E_m = \text{span}\{h_\alpha \otimes e_I : |\alpha| + |I| = m\}$$

are mutually orthogonal, exhaust $\mathcal{H}$, and are invariant under D . The restriction of D to E_m is a self-adjoint matrix with square mI . Its orthogonal direct sum is self-adjoint, with domain $\{\psi : \sum_m m \|\psi_m\|^2 < \infty\}$. Finite sums of the E_m are a core, proving essential self-adjointness. The displayed norm identity follows first on polynomials and then by closure. The Hermite characterization of the closed Gaussian gradient gives the stated Sobolev domain. Finally D changes form parity, which proves the grading relation. $\square$

Proposition I.2 (compact resolvent, summability, and index). The resolvent of D is compact. For every $t > 0$,

$$\text{Tr}(e^{-tD^2}) = \left(\frac{1 + e^{-t}}{1 - e^{-t}} \right)^n, \quad \text{Str}(e^{-tD^2}) = 1.$$

Writing D^+ for its even-to-odd part gives

$$\boxed{\text{ind } D^+ = 1.}$$

Moreover $(1 + D^2)^{-p/2}$ is trace class exactly when $p > 2n$.

Proof. Each E_m is finite dimensional, while $\|(D - i)^{-1}|_{E_m}\| = (m + 1)^{-1/2} \rightarrow 0$, proving compactness. The bosonic trace is $\prod_{j=1}^n \sum_{r \geq 0} e^{-tr} = (1 - e^{-t})^{-n}$. The fermionic trace is $(1 + e^{-t})^n$, while the fermionic supertrace is $(1 - e^{-t})^n$. Their products prove the trace formulas. The sole zero-energy vector is $\Omega = 1 \otimes 1$, an even form. For $m > 0$, D/m is the inverse of D on E_m and swaps parity. Thus every positive-energy eigenspace has matching even and odd dimensions; the odd kernel is zero and the even kernel is $\mathbb{C}\Omega$. The range is closed since the least positive singular value of D is one. This proves the index directly, as well as its heat supertrace formula. Finally

$$\dim E_m = \sum_{q=0}^{\min(n,m)} \binom{n}{q} \binom{n+m-q-1}{n-1}$$

has order m^{n-1} as $m \rightarrow \infty$. Comparing the trace series with $\sum_m m^{n-1-p/2}$ gives the summability assertion. $\square$

The resulting summability dimension is $2n$, the oscillator's phase-space growth dimension. It is not a claim that the underlying Euclidean input space has dimension $2n$, nor that this triple satisfies the reconstruction axioms of a compact n -dimensional spin manifold.

The relation to Bott–Dirac operators. The unitary

$$U : L^2(\gamma_n) \longrightarrow L^2(dx), \quad Uf = (2\pi)^{-n/4} e^{-|x|^2/4} f$$

conjugates D to

$$UDU^{-1} = \sum_j (\varepsilon_j - \iota_j) \partial_j + \frac{1}{2} \sum_j (\varepsilon_j + \iota_j) x_j.$$

Its square is $-\Delta + |x|^2/4 - n/2 + N_f$. This is the harmonic-oscillator/Bott–Dirac construction, with the coordinate and scalar normalizations displayed explicitly. The index-one oscillator is also the basic analytic mechanism appearing in Higson–Kasparov–Trout’s Bott-periodicity work; no infinite-dimensional theorem from that paper is required for the proof above.

3.2 An actual algebra, and the regularity obstruction at a gate

Let $\mathcal{A}_{\text{Lip}}$ be the algebra of bounded Lipschitz complex functions on $\mathbb{R}^n$, represented by multiplication and given norm $\|f\|_\infty + \text{Lip}(f)$.

Proposition I.3 (bounded commutators). Multiplication by $f \in \mathcal{A}_{\text{Lip}}$ preserves $\text{Dom } D$ and

$$[D, M_f] = \sum_j (\varepsilon_j - \iota_j) M_{\partial_j f}.$$

For real f , $\|[D, M_f]\| = \text{ess sup } |\nabla f|$. Thus $(\mathcal{A}_{\text{Lip}}, \mathcal{H}, D, \Gamma)$ is an even, theta-summable spectral triple in the bounded-commutator sense.

Proof. The Sobolev product rule and boundedness of f and its weak gradient show that M_f preserves the domain described in Proposition I.1. Computing the weak first derivatives proves the commutator formula. The Clifford relation

$$\left(\sum_j (\varepsilon_j - \iota_j) v_j \right)^* \left(\sum_j (\varepsilon_j - \iota_j) v_j \right) = |v|^2 I$$

for real v gives the exact norm. Faithfulness follows from the full support of γ_n and continuity of Lipschitz functions; summability follows from Proposition I.2. $\square$

One can use a genuinely noncommutative algebra without changing D . Let $\mathcal{A}_D$ consist of the bounded even operators a such that both a and a^* preserve $\text{Dom } D$ and $[D, a]$ is bounded. Give it the graph norm $\|a\|_D = \|a\| + \|[D, a]\|$. It is a unital Banach star algebra: the Leibniz rule proves closure under products, and completeness follows from the closedness of D by applying convergent operator sequences to each $\psi \in \text{Dom } D$. It contains the multipliers above and every rank-one operator $|\psi\rangle\langle\phi|$ with $\psi, \phi \in \text{Dom } D$ of the same parity, because

$$[D, |\psi\rangle\langle\phi|] = |D\psi\rangle\langle\phi| - |\psi\rangle\langle D\phi|.$$

For example the vacuum projection and $|\Omega\rangle\langle h_1|$, where h_1 is a normalized first Hermite polynomial regarded as an even zero-form, do not commute. Consequently $(\mathcal{A}_D, \mathcal{H}, D, \Gamma)$ is a concrete noncommutative spectral triple. This assertion requires no claim of regularity for iterated commutators with $|D|$; such stronger hypotheses would need separate checks before invoking a local residue index formula.

Proposition I.4 (ReLU and gate multipliers occupy different domains). Let $w \neq 0$ and $r(x) = (w \cdot x + b)_+$. Then:

1. M_r is an unbounded multiplication operator, so r is not an element of the bounded algebra above. Nevertheless $r\Omega \in \text{Dom } D$, and every bounded clipping $r_R = \min(R, r)$ belongs to $\mathcal{A}_{\text{Lip}}$.

2. The gate $g(x) = \mathbf{1}_{\{w \cdot x + b > 0\}}$ is a bounded orthogonal projection in $L^\infty(\gamma_n)$, but $g\Omega \notin \text{Dom } D$. In particular $M_g \notin \mathcal{A}_D$.

Proof. ReLU has at most linear growth and weak gradient $w\mathbf{1}_{\{w \cdot x + b > 0\}}$. Gaussian integrability proves the first domain assertion, and clipping gives a bounded Lipschitz function. For the second assertion rotate coordinates so the boundary is $x_1 = c$. The distributional derivative of its indicator in the first direction is the nonzero surface measure $\delta_{x_1=c}$. It is not represented by a locally square-integrable function. On compact neighborhoods of the boundary, Gaussian and Lebesgue Sobolev regularity are equivalent, so the indicator is not in $W^{1,2}(\gamma_n)$. Apply Proposition I.1. $\square$

Thus an exact gate projection cannot simply be inserted as a smooth projection in this spectral triple's K_0 pairing. A smooth scalar approximation to a gate loses idempotency on its transition region. Introducing a matrix-valued projection, a boundary algebra, or a different Dirac operator is possible, but changes the data and requires a new comparison theorem.

3.3 The index pairing and what it records

Proposition I.5 (bounded Fredholm module). Set $F = D(1 + D^2)^{-1/2}$. Then F is odd and self-adjoint, $F^2 - I$ is compact, and $[F, a]$ is compact for every $a \in \mathcal{A}_D$. Its even-to-odd part has index one.

Proof. The first two claims follow directly from functional calculus and compact resolvent. Here is a compactness proof for the commutator. Put $B = [D, a]$, $b_s = \sqrt{1 + s^2}$, $R_s = (1 + D^2 + s^2)^{-1}$, and $K_\pm = (D \pm ib_s)^{-1}$. The first-order resolvent identity gives

$$DR_s = \frac{1}{2}(K_- + K_+), \quad [DR_s, a] = -\frac{1}{2}(K_-BK_- + K_+BK_+).$$

This uses only preservation of $\text{Dom } D$; preservation of $\text{Dom } D^2$ is not assumed. Each term is compact, and $\|[DR_s, a]\| \leq \|B\|/(1 + s^2)$ because $\|K_\pm\| \leq b_s^{-1}$. The strong functional-calculus integral

$$F = \frac{2}{\pi} \int_0^\infty DR_s ds$$

therefore gives a norm-convergent integral of compact operators for $[F, a]$. On the complement of Ω , the singular values of F are at least $1/\sqrt{2}$; its kernel is $\mathbb{C}\Omega$. Its index is consequently one. $\square$

For an even projection $p \in M_q(\mathcal{A}_D)$, the compression

$$p(F^+ \otimes I_q)p : p(\mathcal{H}^+ \otimes \mathbb{C}^q) \longrightarrow p(\mathcal{H}^- \otimes \mathbb{C}^q)$$

is Fredholm. Its parametrix modulo compact operators is the corresponding compression of F^- . The resulting integer depends on the K_0 class of p . In particular the unit projection pairs to one. This is precisely the even index pairing treated in Connes, Chapter IV, Section 8, Proposition 18 and Theorem 19.

For completeness, the same verified hypotheses support the JLO entire cochain. For $a_j \in \mathcal{A}_D$ and $t > 0$ put

$$\tau_{2k,t}(a_0, \dots, a_{2k}) = t^k \int_{\Delta_{2k}} \text{Str} (a_0 e^{-s_0 t D^2} [D, a_1] e^{-s_1 t D^2} \dots \\ \dots [D, a_{2k}] e^{-s_{2k} t D^2}) ds,$$

where $s_j \geq 0$ and $\sum s_j = 1$. Schatten Holder inequalities give

$$|\tau_{2k,t}(a_0, \dots, a_{2k})| \leq \frac{t^k}{(2k)!} \|a_0\| \prod_{j=1}^{2k} \|[D, a_j]\| \text{Tr}(e^{-t D^2}).$$

Thus every integral and the entire growth condition are controlled. The cocycle and index-pairing assertions used here are Connes' IV.8, Theorems 19, 21, and 22, or Getzler–Szenes, Theorems A

and D , under the hypotheses stated in the source record below. Individual cochain evaluations can vary with analytic data; their pairing with a fixed K_0 class is the index. The zeroth component on the unit is exactly $\tau_{0,t}(1) = 1$.

3.4 A genuine source tangent and its third-chaos spectral support

Let T be a real symmetric third-order tensor in whitened coordinates. Define

$$H_{ijk}(x) = x_i x_j x_k - \delta_{ij} x_k - \delta_{ik} x_j - \delta_{jk} x_i, \quad S_T(x) = \frac{1}{6} \sum_{ijk} T_{ijk} H_{ijk}(x).$$

All tensor sums use ordered indices and the ordinary Frobenius norm.

Proposition I.6 (a positive source path with prescribed third cumulant). The density family

$$\frac{d\mu_\epsilon}{d\gamma_n} = \frac{(1 + \epsilon S_T/2)^2}{1 + \epsilon^2 \|S_T\|_2^2/4}$$

is a probability law for every real ϵ . At zero its score is S_T ; the first variations of its mean and covariance vanish, its third-cumulant variation is T , and every other cumulant has zero first variation. Moreover

$$N_b S_T = 3S_T, \quad \|S_T\|_2^2 = \frac{1}{6} \|T\|_F^2.$$

Proof. S_T is a centered third Hermite polynomial. The square in the numerator is nonnegative, and its Gaussian integral is exactly the denominator. Its derivative at zero is S_T . Gaussian integration by parts, or the Hermite generating function, gives

$$\mathbb{E}[S_T(Z)e^{u \cdot Z}] = \frac{1}{6} T[u, u, u] e^{|u|^2/2}.$$

The family above is a polynomial times a Gaussian, so its moment generating function exists locally, and its log moment generating function has derivative $T[u, u, u]/6$ at zero. Comparing coefficients proves all the cumulant claims. Hermite degree gives the eigenvalue three. Finally Wick contraction gives, writing $(b_1, b_2, b_3) = (a, b, c)$, $\mathbb{E}[H_{ijk} H_{abc}] = \sum_{\pi \in S_3} \delta_{i, b_{\pi(1)}} \delta_{j, b_{\pi(2)}} \delta_{k, b_{\pi(3)}}$. Contracting with symmetric T twice yields $\mathbb{E}[S_T^2] = 6\|T\|_F^2/36$. $\square$

This construction avoids treating the sign-changing density $(1 + \epsilon S_T) d\gamma_n$ as a probability measure. It proves that the first variation is an actual probability-law tangent, without claiming that a third cumulant alone specifies a finite perturbation.

Proposition I.7 (the response is a spectral matrix element). Let $Y = f(Z)$ be a real response with $\mathbb{E}|Y|^8 < \infty$, and put $m = \mathbb{E}Y$, $v = \text{Var } Y$, $c_3 = \kappa_3(Y)$. Define

$$I_f = (Y - m)^4 - 6v(Y - m)^2 - 4c_3(Y - m).$$

Then the source path of Proposition I.6 gives

$$R_T(f) := \left. \frac{d}{d\epsilon} \kappa_4(f(Z_\epsilon)) \right|_0 = \langle S_T, I_f \rangle = \langle S_T, \Pi_3 I_f \rangle,$$

where $\Pi_3 = \mathbf{1}_{\{3\}}(N_b)$ on scalar functions. Equivalently,

$$R_T(f) = \text{Res}_{z=3} \langle S_T, (z - N_b)^{-1} I_f \rangle.$$

This is an ordinary observable-weighted resolvent residue, not a Connes–Moscovici residue index formula. For the normalized fourth gain $g_4 = \kappa_4(Y)/(3v^2)$, $v > 0$, replace I_f by

$$J_f = \frac{I_f}{3v^2} - \frac{2g_4}{v}(Y - m)^2.$$

All the displayed spectral identities then hold for $\delta_T g_4$.

Proof. The first derivative of expectation is the functional $\mathcal{L}_T(A) = \mathbb{E}[S_T A]$, and $\mathcal{L}_T(1) = 0$. Differentiating the centered fourth moment gives $\mathcal{L}_T((Y - m)^4) - 4c_3 \delta m$; differentiating $-3v^2$ gives $-6v \delta v$. Here $\delta m = \mathcal{L}_T(Y - m)$ and $\delta v = \mathcal{L}_T((Y - m)^2)$, proving the influence formula. The integrability assumption justifies differentiation in the polynomial density family by Cauchy–Schwarz. It also gives $I_f \in L^2(\gamma_n)$. Self-adjointness and $S_T \in \text{ran } \Pi_3$ prove the projection identity. Expanding the resolvent in Hermite eigenspaces gives

$$\langle S_T, (z - N_b)^{-1} I_f \rangle = \frac{\langle S_T, I_f \rangle}{z - 3},$$

which proves the residue formula. The derivative of $\kappa_4/(3v^2)$ is $\delta \kappa_4/(3v^2) - 2g_4 \delta v/v$, giving J_f . $\square$

The creation-operator description of the same identity is

$$S_T = \frac{1}{6} \sum_{ijk} T_{ijk} a_i^* a_j^* a_k^* \Omega, \quad R_T(f) = \langle S_T, M_{I_f} \Omega \rangle.$$

The creation and annihilation operators genuinely do not commute. For a finite ReLU network and a fixed scalar output, polynomial growth gives the required Gaussian moments. The formula therefore applies to its exact scalar response at a Gaussian reference law. It does not assert that a Gaussian reference accurately describes the actual deep-layer law.

What must be translated back to the ledger. An incoming unit third-gain perturbation must first be translated into its actual whitened tensor T . A coherent output gain is a chosen contraction or fitted coordinate, so its full differential, including moving templates, must replace J_f when necessary. To isolate omitted sectors, apply the same response functional to the difference between the full readout and the retained-sector readout, with the same reference law and coordinate frame. The index calculation supplies none of these identifications automatically.

3.5 Heat quantities retain information that the index cancels

Let $P_t = e^{-tN_b}$ and $q = e^{-t}$. Hermite generating functions show that

$$(P_t f)(x) = \mathbb{E} f(qx + \sqrt{1 - q^2} Z'),$$

where Z' is an independent standard Gaussian. This is the Mehler semigroup. The formula can be proved first on Hermite polynomials and extended to $L^2(\gamma_n)$ using contraction and density.

Proposition I.8 (inserted heat supertrace). For every bounded measurable scalar f and $t > 0$,

$$\boxed{\text{Str}(M_f e^{-tD^2}) = \mathbb{E} f\left(\sqrt{\coth(t/2)} Z\right)}.$$

In particular the limit as $t \rightarrow \infty$ is $\mathbb{E} f(Z)$ for bounded continuous f . With $\sigma_t = \sqrt{\coth(t/2)}$, the rescaled insertion $f_t(x) = f(x/\sigma_t)$ gives

$$\text{Str}(M_{f_t} e^{-tD^2}) = \mathbb{E} f(Z)$$

at every $t > 0$.

Proof. The Mehler kernel relative to γ_n is

$$k_t(x, y) = (1 - q^2)^{-n/2} \exp\left(\frac{2q x \cdot y - q^2(|x|^2 + |y|^2)}{2(1 - q^2)}\right).$$

Since M_f acts trivially on the exterior factor,

$$\text{Str}(M_f e^{-tD^2}) = (1 - q)^n \int f(x) k_t(x, x) d\gamma_n(x).$$

On the diagonal, $k_t(x, x) = (1 - q^2)^{-n/2} e^{q|x|^2/(1+q)}$. The measure $(1 - q)^n k_t(x, x) d\gamma_n(x)$ is exactly the Gaussian law with covariance $(1 + q)(1 - q)^{-1}I = \coth(t/2)I$. This proves both identities. The trace computation is justified by the absolutely convergent Hermite trace series, bounded by $\|f\|_\infty(1 - q)^{-n}$ before the fermionic factor. The limit follows by dominated convergence after representing the integral with a fixed Z . $\square$

For polynomial-growth insertions, the same identity is defined rigorously by the heat sandwich $e^{-tD^2/2} M_f e^{-tD^2/2}$: its absolute trace is controlled by the finite Gaussian integral with $|f|$. This includes the polynomial-growth moment and influence insertions of a finite ReLU network. Such inserted traces encode an expectation, and so do not remove its integration cost. Their significance is that the spectral framework contains the required analytic information even though its index discards that information.

For the third source specifically, Proposition I.7 also gives the exact identity and bound

$$R_T(f) = e^{3t} \langle S_T, P_t I_f \rangle, \quad |R_T(f)| \leq \frac{\|T\|_F}{\sqrt{6}} \|\Pi_3 I_f\|_2.$$

The factor e^{3t} is essential; smoothing alone is not a free reduction of the response or its statistical uncertainty.

The index cancels every positive-energy even/odd pair. The source S_T is an even zero-form of bosonic energy three, so its contribution belongs exactly to a sector canceled by that index. This is a direct explanation, within this construction, of why the index and the susceptibility answer different questions. The surviving vacuum is a constant Gaussian zero-form; it has not been identified with the network's random-scale transport mode.

3.6 A precise limitation of index-only formulas

Proposition I.9 (homotopy limitation). Suppose a connected real parameter interval J gives a norm-continuous family of bounded Fredholm operators A_t . Then $\text{ind } A_t$ is constant. Consequently a nonconstant continuous response on J cannot be a function only of this integer index. The same conclusion holds for finitely many indices of such families.

Proof. Fix t_0 and split the domain into $\ker A_{t_0}$ and its orthogonal complement, and the codomain into $\text{ran } A_{t_0}$ and its finite dimensional orthogonal complement. The block from the infinite dimensional domain complement to the range is invertible. It remains invertible under a sufficiently small norm perturbation. Block Gaussian elimination then reduces A_t to this invertible block plus a map between the fixed finite dimensional kernel and cokernel spaces. The latter has index equal to the dimension of its domain minus the dimension of its codomain, independent of its entries. Thus the Fredholm index is locally constant, and hence constant on J . The final assertions follow. $\square$

This is a limitation on using integer homotopy invariants alone. It is not an impossibility theorem for noncommutative geometry, spectral analysis, analytic cyclic cochains, or a representation using more than an index.

Scope concerning measured and semifinite indices. Proposition I.9 concerns ordinary Fredholm indices and, below, finite projection compressions. A specified semifinite operator algebra can instead carry a trace-valued index. At the algebraic level, if an index class $\xi \in K_0(\mathcal{J})$ is invariant along a deformation and a fixed trace induces $\tau_* : K_0(\mathcal{J}) \rightarrow \mathbb{R}$, then $\tau_*(\xi)$ is invariant along that deformation as well. This is immediate by applying the fixed map τ_* to the constant class. Allowing the trace or a transverse measure to vary can produce real-valued variation; that observation is not excluded by our ordinary-index example. Using it for the network would require an explicit algebra, trace, Fredholm class, and a proved comparison between its trace pairing and the desired cumulant response. None is supplied by the Gaussian index calculation alone.

Proposition I.10 (a bias-free ReLU family with a varying normalized response). Let Z_1, Z_2 be independent standard Gaussians and

$$Y_a = (Z_1)_+ + aZ_2 = (Z_1)_+ + a(Z_2)_+ - a(-Z_2)_+, \quad a \geq 0.$$

This is a scalar linear readout of a bias-free ReLU layer. Use the fixed third source $T = e_1^{\otimes 3}$, so $S_T = (Z_1^3 - 3Z_1)/6$. Set

$$u = (2\pi)^{-1/2}, \quad v_0 = \frac{1}{2} - u^2, \quad c_4 = \frac{3}{4} - 2u^2 - 6u^4, \quad C = u + 4u^3.$$

Then the normalized response satisfies

$$\boxed{\delta_T \left(\frac{\kappa_4(Y_a)}{3 \text{Var}(Y_a)^2} \right) = \frac{C(v_0 + a^2) - (2/3)c_4u}{3(v_0 + a^2)^3}.$$

It is positive at $a = 0$, where it equals $7u^3/(9v_0^3)$, and tends to zero as $a \rightarrow \infty$. The Gaussian spectral triple is the same for every a and has index one throughout.

Proof. The half-Gaussian moments of $(Z_1)_+$ are $u, 1/2, 2u, 3/2$ in orders one through four. They give v_0 and c_4 above. Direct integration of $S_T(Z_1)(Z_1)_+^k$, for $k = 1, 2, 3, 4$, gives respectively $0, u/3, 1/2, 4u$. Therefore $\delta m = 0$, $\delta v = u/3$, and

$$\delta \kappa_4 = 4u - 4u(1/2) - 6(1/2)(u/3) + 12u^2(u/3) = u + 4u^3 = C.$$

The perturbed density depends only on Z_1 , so independence of Z_2 persists along the source path. Adding aZ_2 adds a^2 to the variance and zero to the fourth cumulant, without changing their source derivatives. Differentiating the normalized ratio gives the displayed expression. At $a = 0$, its numerator simplifies to $7u^3/3$ before division by $3v_0^3$; the limit at infinity is immediate. The operator D depends only on the fixed two-dimensional Gaussian reference, proving the index assertion. $\square$

This example concerns a well-defined unit third-cumulant source and normalized output gain. It does not assert that this is the exact coordinate convention of Note XIX. It establishes that even within bias-free ReLU readouts the quantitative response is not determined by the Gaussian index.

The source state is not automatically a cyclic trace. The Gaussian vacuum functional on the noncommutative algebra is $\omega_0(a) = \langle \Omega, a\Omega \rangle$. It is not tracial. Let h_1 be a normalized first Hermite polynomial regarded as an even zero-form and put $A = |\Omega\rangle\langle h_1| \in \mathcal{A}_D$. Then

$$\omega_0(AA^*) = 1, \quad \omega_0(A^*A) = 0.$$

Both equalities follow by multiplying the two rank-one operators. Thus the source-readout matrix element is not automatically a cyclic index pairing. On the commutative multiplication algebra, the same functional is the Gaussian integral, which is of course tracial there.

A faithful type-I state can be specified by choosing a thermal parameter $\beta > 0$ and setting

$$\rho_\beta = \frac{e^{-\beta D^2}}{\text{Tr}(e^{-\beta D^2})}, \quad \omega_\beta(a) = \text{Tr}(\rho_\beta a).$$

The density is positive, injective, and trace class, so ω_β is a faithful normal state on $\mathcal{B}(\mathcal{H})$. The type-I modular formula, under exactly these hypotheses, is

$$\sigma_s^{\omega_\beta}(a) = \rho_\beta^{is} a \rho_\beta^{-is} = e^{-i\beta s D^2} a e^{i\beta s D^2}.$$

For a direct verification, realize its GNS Hilbert space as Hilbert–Schmidt operators, with cyclic separating vector $\rho_\beta^{1/2}$ and left multiplication representation. If $\rho_\beta e_i = r_i e_i$, the Tomita operator sends the Hilbert–Schmidt matrix unit E_{ij} to $\sqrt{r_i/r_j} E_{ji}$. Its modular operator consequently satisfies $\Delta E_{ij} = (r_i/r_j) E_{ij}$, which gives $\Delta^{is} L_a \Delta^{-is} = L_{\rho_\beta^{is} a \rho_\beta^{-is}}$. The finite matrix units form a core for the diagonal modular operator; the displayed unitary conjugations extend normally to all bounded operators. This is the explicit type-I instance of the Tomita formula in Connes, Chapter I, Section 3. It is an actual modular dynamics, distinct from the Mehler heat semigroup. For example, in one coordinate and for $\xi \neq 0$,

$$P_t e^{i\xi x} = e^{-\frac{1}{2}(1-e^{-2t})\xi^2} e^{ie^{-t}\xi x},$$

which maps a unitary multiplier to a function of modulus less than one. An automorphism preserves unitaries. The Mehler formula therefore cannot be the modular automorphism group. No equality between this chosen thermal state and a fitted network state is asserted.

Proposition I.11 (Fredholm pairs do not replace alignment data). For orthogonal projections P, Q with $P - Q$ compact, the operator $QP : P\mathcal{H} \rightarrow Q\mathcal{H}$ is Fredholm and

$$\text{ind}(QP) = \dim(P\mathcal{H} \cap \ker Q) - \dim(Q\mathcal{H} \cap \ker P).$$

If P and Q have finite ranks r, s , this index is always $r - s$, independently of their relative angles.

Proof. PQ is a parametrix: $PQP - P = P(Q - P)P$ and $QPQ - Q = Q(P - Q)Q$ are compact. The kernel is $P\mathcal{H} \cap \ker Q$, and the kernel of the adjoint is $Q\mathcal{H} \cap \ker P$. In finite dimensions, rank-nullity gives $r - s$. $\square$

For example, in $\mathbb{R}^2$, take P onto e_1 and Q_θ onto $\cos \theta e_1 + \sin \theta e_2$. The index is zero for every θ , while the sole singular value of $Q_\theta P$ is $|\cos \theta|$. Thus a projection-pair index can record a dimension imbalance, but the singular values and angles are the quantities that measure retained versus lost response. A finite two-gain fit does not establish a Fredholm-pair description of the full neural continuation problem.

3.7 A spectral gate diagnostic with exact regularity thresholds

Proposition I.12 (Gaussian gate noise sensitivity). Let $g(x) = \mathbf{1}_{\{x_1 > 0\}}$ and $P_t = e^{-tN_b}$. Then

$$\langle g, (I - P_t)g \rangle = \frac{\arccos(e^{-t})}{2\pi}.$$

Consequently, for $0 < s < 1$,

$$g \in \text{Dom}(N_b^{s/2}) \iff s < \frac{1}{2}.$$

The scalar ReLU $r(x) = (x_1)_+$ belongs to $\text{Dom}(N_b^{s/2})$ exactly for $0 \leq s < 3/2$.

Proof. Under the Mehler coupling, X_1 and its time- t partner are standard Gaussians of correlation $\rho = e^{-t}$. Write them as U and $\cos \theta U + \sin \theta V$, where U, V are independent standard Gaussians and

$\theta = \arccos \rho$. The angle of (U, V) is uniform. Their signs disagree on sectors with total angle 2θ , so the disagreement probability is θ/π . Stationarity and $g^2 = g$ give $\langle g, (I - P_t)g \rangle = \frac{1}{2}\mathbb{E}(g(X) - g(X_t))^2$. This proves the identity.

For $0 < s < 1$, interpret the energy below as the extended spectral quadratic form $\int \lambda^s d\langle g, E_\lambda g \rangle$, with value $+\infty$ outside $\text{Dom}(N_b^{s/2})$. Spectral calculus and Tonelli's theorem give

$$\langle g, N_b^s g \rangle = \frac{s}{\Gamma(1-s)} \int_0^\infty t^{-1-s} \langle g, (I - P_t)g \rangle dt.$$

Indeed the scalar identity is $\lambda^s = \frac{s}{\Gamma(1-s)} \int_0^\infty (1 - e^{-t\lambda}) t^{-1-s} dt$, obtained by integration by parts and scaling. The large- t integral is finite. As $t \downarrow 0$ the gate energy is asymptotic to $\sqrt{2t}/(2\pi)$, so the small- t integral is finite exactly when $s < 1/2$.

In one dimension write $r = \sum_{m \geq 0} r_m h_m$ and $g = \sum_{m \geq 0} g_m h_m$. The weak derivative relation $\partial r = g$ and the Hermite ladder identity give $g_m = \sqrt{m+1} r_{m+1}$. For $s > 1$, convergence of $\sum m^s |r_m|^2$ is consequently equivalent to convergence of $\sum (m+1)^{s-1} |g_m|^2$. The gate result gives the threshold $s < 3/2$ for $1 < s < 2$, including failure at $3/2$. ReLU is already in the first Gaussian Sobolev space; nesting of the domains gives the result for all $s \geq 0$. The other coordinates contribute only their vacuum factors. $\square$

These are quantities in the same oscillator calculus as the index, but they measure actual gate sensitivity and regularity. In particular a gate can be perfectly bounded while carrying infinitely many non-negligible high-energy components. Finite ReLU influence functions have at least first Gaussian Sobolev regularity because they are continuous, piecewise polynomial, and have polynomially bounded weak gradients. Exponential spectral convergence does not follow from that fact.

3.8 The quantitative object for a contracted source ledger

For fixed influence functions $I_1, \dots, I_m \in L^2(\gamma_n)$ define

$$B : \text{ran } \Pi_3 \longrightarrow \mathbb{R}^m, \quad (Bs)_j = \langle s, I_j \rangle.$$

This is the exact linear response operator from Gaussian third-chaos source scores to the selected output queries. Its Gram matrix is

$$(BB^*)_{ij} = \langle \Pi_3 I_i, \Pi_3 I_j \rangle.$$

These singular values, rather than a Fredholm index, determine whether all the selected responses admit a common low-rank approximation. The best rank- r operator approximation in operator norm has error $\sigma_{r+1}(B)$: truncating a singular-value decomposition attains this error, while any rank- r operator has a nonzero kernel vector on the span of the first $r+1$ right singular vectors, proving the matching lower bound. This is a finite statement about a specified source and query space, not a claim that those spaces are cheap to construct.

For one tensor source and one approximation $\hat{I}$,

$$|\langle S_T, I - \hat{I} \rangle| \leq \frac{\|T\|_F}{\sqrt{6}} \|\Pi_3(I - \hat{I})\|_2.$$

The proof is the projection identity and Cauchy-Schwarz. If $T = \sum_{r=1}^R \lambda_r v_r^{\otimes 3}$ is already supplied in a factored form, its score can be evaluated as

$$S_T(x) = \frac{1}{6} \sum_{r=1}^R \lambda_r ((v_r \cdot x)^3 - 3\|v_r\|^2(v_r \cdot x)),$$

using $O(nR)$ arithmetic per sample. No assertion of small tensor rank is made when such a factorization is not available.

At successive layers, the relevant query space must additionally be closed under backward transport, or supplemented with the observable memory left by failure of that closure. The index calculations do not establish that property. Their useful contribution is a rigorous spectral setting, an exact distinction between topological and analytic data, and a concrete third-chaos response operator to which low-rank and error questions can be addressed.

3.9 Checked primary sources and the source-to-target dictionary

1. Alain Connes, *Noncommutative Geometry*, Academic Press, 1994, author-hosted copy: <https://alainconnes.org/wp-content/uploads/book94bigpdf.pdf>. Checked Chapter IV, Section 8, Proposition 18 and Theorem 19 (printed page 414), and Theorems 21–22 (printed pages 416–417). Chapter I, Section 3 (printed page 49) supplies the modular-operator convention. The required data are an even theta-summable K-cycle over a locally convex algebra, bounded commutators continuous in the algebra topology, and a matrix projection for the K_0 pairing. They supply the index pairing and the JLO entire cocycle. Our $\mathcal{A}_D$ has its graph Banach norm, Γ is form parity, and the heat bound is Proposition I.2.
2. Ezra Getzler and Andras Szenes, *On the Chern character of a theta-summable Fredholm module*, author-hosted manuscript: <https://bpb-us-e1.wpmucdn.com/sites.northwestern.edu/dist/c/2278/files/2019/08/char.pdf>. Checked the defining hypotheses on page 1, Theorem A on page 2, and Theorems B–D on page 3. These require a unital Banach algebra represented evenly, an odd self-adjoint D , a uniform graph-norm commutator bound, and a slightly stronger-than-unit-time heat-trace condition. Our all- t trace formula satisfies the latter. Theorem A gives the cocycle and specified homotopy invariance; Theorem D gives the index pairing. Their chain normalizations differ from Connes', so no coefficients are transferred between conventions.
3. Nigel Higson, Gennadi Kasparov, and Jody Trout, *A Bott Periodicity Theorem for Infinite Dimensional Euclidean Space*, *Advances in Mathematics* **135** (1998), 1–40, <https://doi.org/10.1006/aima.1997.1706>. Publication identity was authenticated through the publisher and Jody Trout's publication list, <https://math.dartmouth.edu/~jodyt/publ.htm>. The publisher abstract identifies its index-one elliptic operator. The full theorem was not used: the finite-dimensional oscillator, compactness, and index needed here were proved explicitly in Propositions I.1–I.2.

The checked source statements establish the spectral-triple and index-theory machinery under its hypotheses. The Gaussian source path, cumulant influence formula, explicit ReLU family, gate regularity calculation, and query response bounds above are independent finite derivations. No novelty claim is made.

4 The exact radial Ward identity

This section replaces an assumed unit row sum by an exact identity for a specified positive random-scale family. The choice of family is part of the statement.

Let $\xi \sim N(0, 1)$ be independent of a random vector Z , and put

$$G_t = \exp\left\{\frac{1}{2}\sqrt{t}\xi - \frac{1}{4}t\right\}, \quad t \geq 0.$$

Then $\mathbb{E}G_t^2 = 1$, $\text{Var}(G_t^2) = e^t - 1$, and

$$\mathbb{E}G_t^k = e^{tk(k-2)/8}. \quad (37)$$

Thus differentiation at $t = 0$ is also differentiation with respect to the mixture gain $g = \text{Var}(G_t^2)$ at $g = 0$.

Theorem 4.1 (Radial generator and exact intertwining). *Write $\mathcal{E} = \sum_i z_i \partial_i$. For a smooth test function whose derivatives through order three have polynomial growth, the infinitesimal scale operator is*

$$\mathcal{S}f = \frac{1}{8}(\mathcal{E}^2 - 2\mathcal{E})f = \frac{1}{8} \left(\sum_{ij} z_i z_j \partial_{ij} f - \sum_i z_i \partial_i f \right).$$

If $F(cz) = cF(z)$ for every $c > 0$, then the scale averaging operators obey

$$\mathbb{E}_\xi[f(G_t F(z))] = \mathbb{E}_\xi[(f \circ F)(G_t z)].$$

Consequently the radial generators intertwine wherever their derivatives exist. The averaging identity itself requires no smoothness of F and applies to bias-free ReLU networks.

Proof. The log-radius increment has mean $-t/4$ and variance $t/4$, so its generator is $\frac{1}{8}\partial_r^2 - \frac{1}{4}\partial_r$. Under $z \mapsto e^r z$, $\partial_r = \mathcal{E}$. Alternatively, a Taylor expansion using $\mathbb{E}(G_t - 1) = -t/8 + o(t)$ and $\mathbb{E}(G_t - 1)^2 = t/4 + o(t)$ gives the displayed formula. Polynomial moment bounds justify the expansion for the stated tests. The averaging identity follows pointwise from $F(G_t z) = G_t F(z)$. Differentiation gives intertwining without differentiating across a ReLU gate: positive radial scaling never changes a gate sign. $\square$

Theorem 4.2 (Scale derivatives of cumulants). *Let Y have finite moments through order r , independently of the Gaussian ξ defining G_t , and let κ_j be its cumulants. If $Y_t = G_t Y$, then, for $2 \leq n \leq r$,*

$$\dot{\kappa}_n = \frac{1}{8} \left[n(n-2)\kappa_n + n(n-1) \sum_{j=0}^{n-2} \binom{n-2}{j} \kappa_{j+1} \kappa_{n-1-j} \right], \quad (38)$$

and $\dot{\kappa}_1 = -\kappa_1/8$. In particular, with $m = \mathbb{E}Y$ and $v = \text{Var} Y$,

$$\dot{m} = -m/8, \quad \dot{v} = m^2/4, \quad (39)$$

$$\dot{\kappa}_3 = \frac{3}{2}mv + \frac{3}{8}\kappa_3, \quad \dot{\kappa}_4 = 3v^2 + 3m\kappa_3 + \kappa_4. \quad (40)$$

Proof. Equation (37) gives $\dot{m}_k = k(k-2)m_k/8$ for each raw moment separately. Cumulants through order r are polynomial functions of those moments, so differentiation is legitimate without existence of a moment-generating function. To organize the finite polynomial calculation, use formal series $M(s) = \sum_{k=0}^r m_k s^k/k!$ and $K(s) = \log M(s)$ modulo s^{r+1} . Then

$$\dot{K}(s) = \frac{1}{8} \{s^2[K''(s) + K'(s)^2] - sK'(s)\} \pmod{s^{r+1}}.$$

Coefficient extraction gives (38). This is a formal finite-order identity; lognormal mixtures generally do not possess a finite real moment-generating function away from zero. $\square$

Corollary 4.3 (Gaussian input scale tangent has no higher first-order cumulants). *Suppose $Z \sim N(\mu, \Sigma)$ and Σ is positive definite. The tangent of $G_t Z$ at zero decomposes into*

$$\dot{\mu}_i = -\mu_i/8, \quad \dot{\Sigma}_{ij} = \mu_i \mu_j / 4, \quad (41)$$

$$T_{ijk}^{(3)} = \frac{1}{2}(\mu_i \Sigma_{jk} + \mu_j \Sigma_{ik} + \mu_k \Sigma_{ij}), \quad (42)$$

$$T_{ijkl}^{(4)} = \Sigma_{ij} \Sigma_{kl} + \Sigma_{ik} \Sigma_{jl} + \Sigma_{il} \Sigma_{jk}. \quad (43)$$

All source cumulants of order $k \geq 5$ have zero first derivative for this family.

Proof. Apply Theorem 4.2 to every scalar projection $a^\top Z$ and polarize the resulting homogeneous polynomials in a . Gaussian cumulants above two vanish. In the quadratic cumulant product in (38), both indices can be at most two only when $n \leq 4$. Equivalently, putting $q = \mu \cdot a$ and $r = a^\top \Sigma a$, the formal derivative of the Gaussian cumulant series is $(-q + q^2 + 2qr + r^2)/8$. Its degree is at most four. $\square$

This corollary concerns the explicitly specified regular scale family at a Gaussian input reference. It does not assert that a rectified Gaussian is Gaussian, that finite gains have no higher cumulants, or that a different rare-event scale perturbation has the same tangent.

Theorem 4.4 (Complete fourth-channel Ward identity). *Let $Y = F(Z)$ be a scalar positively homogeneous readout of polynomial growth, $Z \sim N(\mu, \Sigma)$ with Σ positive definite, and let the scale G_t be independent of Z . Let $\mathcal{L}_3$ and $\mathcal{L}_4$ denote the pure third- and fourth-cumulant Gaussian tangent operators in the directions of Corollary 4.3. Then*

$$D_{\mu, \Sigma} \kappa_4(F(Z))[-\mu/8, \mu\mu^\top/4] + \mathcal{L}_3 \kappa_4(F(Z)) + \mathcal{L}_4 \kappa_4(F(Z)) = 3v^2 + 3m\kappa_3(Y) + \kappa_4(Y). \quad (44)$$

Here the first term is a directional derivative of the reference mean and covariance, and m, v are the output mean and variance at the unperturbed reference. All terms use the same fixed readout.

Proof. Corollary 4.3 decomposes the Gaussian input law tangent into the four displayed cumulant directions. This decomposition is valid as a weak density tangent on polynomial-growth tests: Gaussian integration by parts gives, for a smooth test f ,

$$\mathbb{E} S f(Z) = \frac{1}{8} \mathbb{E} [-\mu \cdot \nabla f + (\mu \cdot \nabla)^2 f + 2(\mu \cdot \nabla) \Delta_\Sigma f + \Delta_\Sigma^2 f], \quad \Delta_\Sigma = \sum_{ij} \Sigma_{ij} \partial_{ij}.$$

The terms are respectively the mean, covariance, third-cumulant, and fourth-cumulant source tangents, with the tensor normalizations stated above. Gaussian score integration extends the identity to polynomial-growth nonsmooth readouts such as ReLU products. Apply the tangent to the influence function of the fourth cumulant. Theorem 4.1 identifies this source perturbation with scaling Y , and Theorem 4.2 evaluates its derivative. $\square$

Normalized gains. For $v > 0$, $g_4 = \kappa_4/(3v^2)$ has scale derivative

$$\dot{g}_4 = 1 + \frac{m\kappa_3}{v^2} + g_4 - \frac{m^2}{2v} g_4. \quad (45)$$

For $m \neq 0$ and $g_3 = \kappa_3/(\frac{3}{2}mv)$,

$$\dot{g}_3 = 1 + \left(\frac{1}{2} - \frac{m^2}{4v} \right) g_3.$$

These follow by the quotient rule. Both derivatives equal one at a Gaussian scalar reference where both gains are defined, but the output of a ReLU layer generally is not such a reference. Thus an imposed unit row sum is not the complete Ward identity. Equation (44), including reference drift and the correct output normalization, is an exact check available to the ledger.

Cheap full scale scores. Write $x = \Sigma^{-1/2}(Z - \mu)$, $a = \Sigma^{-1/2}\mu$, and $r = \|x\|^2$, using a fixed symmetric positive square root. The four orthogonal-chaos components of the radial source score are

$$S_1 = -a \cdot x/8, \quad S_2 = ((a \cdot x)^2 - \|a\|^2)/8, \quad (46)$$

$$S_3 = (a \cdot x)(r - n - 2)/4, \quad S_4 = (r^2 - 2(n + 2)r + n(n + 2))/8. \quad (47)$$

Gaussian Hermite contraction proves these formulas, or they follow by integrating the four differential operators in the previous proof by parts. After whitening, each score costs $O(n)$ per sample. These are the *full* scale tangents. Replacing the ledger's retained input slices by these scores changes the source model unless that replacement is explicitly intended.

5 Compression calculus and moving retained frames

The algebraic use of noncommutativity can be made exact without asserting that the neuron coordinates themselves fail to commute.

Theorem 5.1 (Multiplication and Schwarz defects of compression). *Let P be an orthogonal projection on a Hilbert space, $Q = I - P$, and $\Phi(A) = PAP$ viewed as an operator on $P\mathcal{H}$. For bounded A, B ,*

$$\Phi(AB) - \Phi(A)\Phi(B) = PAQBP, \quad (48)$$

$$\Phi(A^*A) - \Phi(A)^*\Phi(A) = PA^*QAP \succeq 0. \quad (49)$$

If A and B commute, their compressions satisfy

$$[\Phi(A), \Phi(B)] = PBQAP - PAQBP.$$

The Schwarz defect vanishes if and only if $QAP = 0$. The defects for both A and A^* vanish if and only if A commutes with P .

Proof. Insert $I = P + Q$ between the two factors. For the positivity statement, the remainder equals $(QAP)^*(QAP)$. Its vanishing is equivalent to $QAP = 0$. Applying the same argument to A^* gives $PAQ = 0$, and the two off-diagonal blocks vanish exactly when $AP = PA$. The commutator formula follows by subtracting the two multiplication-defect formulas and using $AB = BA$. $\square$

The map Φ is unital and completely positive as a map to $B(P\mathcal{H})$: at every matrix level its positivity follows by compression with $I \otimes P$. Thus the positive defect is an exact operator-theoretic measure of discarded intermediate propagation. If P is finite rank, $F_P = 2P - I$ has compact commutators with all bounded operators. Its ordinary finite compression index is nevertheless blind to the size of these defects. The nonzero singular values, rather than their signed kernel count, measure the approximation error. An index-pattern mask on a fourth tensor should not be identified with this Hilbert projection without constructing that identification.

Theorem 5.2 (Connection of a moving retained subspace). *Let $\theta \mapsto P(\theta)$ be a twice continuously differentiable, finite-rank orthogonal projection on a fixed Hilbert space, with constant rank on its parameter domain. The connection $\nabla s = Pds$ on sections satisfying $Ps = s$ has curvature*

$$\nabla^2 s = P(dP) \wedge (dP)s.$$

If $J(\theta) : \mathbb{C}^r \rightarrow \mathcal{H}$ is a smooth orthonormal frame, $J^*J = I$ and $JJ^* = P$, then the connection one-form is $\mathcal{A} = J^*dJ$ and its curvature is

$$d\mathcal{A} + \mathcal{A} \wedge \mathcal{A} = dJ^*(I - P) \wedge dJ.$$

Proof. Differentiating $P^2 = P$ gives $P(dP)P = 0$. For $Ps = s$, differentiating gives $ds = (dP)s + Pds$. Therefore $Pd(Pds) = P(dP) \wedge ds = P(dP) \wedge (dP)s$. In a frame, $\nabla(Jv) = J(dv + J^*dJv)$. Differentiation gives $d\mathcal{A} = dJ^* \wedge dJ$, while $dJ^*J = -J^*dJ$ implies $dJ^*P \wedge dJ = -\mathcal{A} \wedge \mathcal{A}$. Subtraction proves the second identity. $\square$

This is the Grassmann connection used for projective modules in noncommutative geometry. Here its immediate content is the derivative of a changing projection or gain template. Along a single depth interval there is no two-dimensional cycle on which a Chern number could supply the missing scalar amplitude. More strongly, if the retained vectors $\phi_1(\theta), \dots, \phi_r(\theta)$ are a globally defined independent list, their Gram matrix $G(\theta)$ gives a global orthonormal frame $J = \phi G^{-1/2}$. That bundle is explicitly trivial. Local curvature can still be nonzero, but no missing global frame follows from its mere presence.

Gain coefficients in a moving template. Let $b(\theta)$ be a nonzero template, $R(\theta)$ a symmetric fitting weight, and $d = b^\top Rb > 0$. For

$$g(\theta) = \frac{b^\top RK}{b^\top Rb},$$

direct differentiation gives

$$\delta g = \frac{b^\top R \delta K + (\delta b)^\top R(K - 2gb) + b^\top (\delta R)(K - gb)}{d}.$$

This is the numerical-coordinate counterpart of keeping track of a moving frame. Omitting these terms changes the quantity differentiated; no index theorem repairs an incorrect coordinate derivative.

6 An exact operator criterion for retained-observable closure

6.1 Reference laws, tangent scores, and the direction of the operators

Fix a finite depth L , measurable state spaces E_ℓ , probability laws μ_ℓ , and measurable layer maps $F_\ell : E_\ell \rightarrow E_{\ell+1}$, for $0 \leq \ell < L$. The principal setting of this section assumes

$$\mu_{\ell+1} = (F_\ell)_\# \mu_\ell.$$

Put $\mathcal{H}_\ell = L_0^2(\mu_\ell)$, the real Hilbert space of square-integrable mean-zero functions, and define the backward observable operator

$$U_\ell : \mathcal{H}_{\ell+1} \longrightarrow \mathcal{H}_\ell, \quad U_\ell f = f \circ F_\ell.$$

It is an isometry:

$$\|U_\ell f\|_{\mu_\ell}^2 = \int |f|^2 d(F_\ell)_\# \mu_\ell = \|f\|_{\mu_{\ell+1}}^2.$$

Its adjoint $T_\ell = U_\ell^*$ is the forward score operator. If a regular probability path through μ_ℓ has score s_ℓ , the score of its pushforward is $s_{\ell+1} = T_\ell s_\ell$; equivalently,

$$\langle f, s_{\ell+1} \rangle_{\mu_{\ell+1}} = \langle U_\ell f, s_\ell \rangle_{\mu_\ell}.$$

Conditional expectation gives the same operator, interpreted as a function of $F_\ell(H_\ell)$.

For a differentiable statistic $a(\mu)$, let $\phi_a \in L_0^2(\mu)$ be its influence function:

$$\delta_s a = \langle \phi_a, s \rangle_\mu.$$

Thus the statements below concern exact expectations of fixed observables, or the exact first variation of differentiable moment, cumulant, and gain coordinates at the specified reference laws. They do not, without remainder bounds, assert an exact nonlinear finite-amplitude gain recursion. Fourth-cumulant influence functions require the appropriate eighth moments to lie in L^2 ; gain denominators and fitted-template Gram matrices must also be nondegenerate. A finite fixed ReLU network with a Gaussian source has the required polynomial moments, but division by a zero coordinate normalizer must still be avoided.

Allowed perturbations. All universal quantifiers over scores below are relative to a declared closed tangent space. The full choice $\mathcal{H}_\ell = L_0^2(\mu_\ell)$ tests closure against all square-integrable source variations. If only a closed subspace $\mathcal{S}_\ell$ is allowed, with $T_\ell \mathcal{S}_\ell \subseteq \mathcal{S}_{\ell+1}$, replace each observable by its orthogonal projection onto $\mathcal{S}_\ell$, and replace U_ℓ by the adjoint of $T_\ell|_{\mathcal{S}_\ell}$. This effective backward operator is bounded, but need not be an isometry. The general bounded-operator versions of the closure and memory results remain valid. The simplified identity $I - C_\ell^* C_\ell = E_\ell^* E_\ell$, and the dimension and principal-angle assertions explicitly stated for the full isometric setting below, require that isometry and cannot be transferred to a noninjective effective operator.

This distinction is material. Resetting a Gaussian reference independently at every layer generally does not satisfy the pushforward hypothesis. In particular, a rectified law can have an atom where the Gaussian reference has measure zero; composition on Gaussian L^2 equivalence classes can then fail even to be well-defined. A local Gaussian score calculation remains legitimate on its own specified reference law, but cannot silently be substituted for the exact-law Hilbert operator above.

6.2 Closure, coefficient error, and intrinsic projection leakage

Let $V_\ell \subset \mathcal{H}_\ell$ be a finite-dimensional retained query space. Choose an orthonormal synthesis map $J_\ell : \mathbb{R}^{p_\ell} \rightarrow \mathcal{H}_\ell$, with range $J_\ell = V_\ell$, and write

$$P_\ell = J_\ell J_\ell^*, \quad Q_\ell = I - P_\ell, \quad C_\ell = J_\ell^* U_\ell J_{\ell+1}, \quad E_\ell = Q_\ell U_\ell J_{\ell+1}.$$

The retained score coordinates are $x_\ell = J_\ell^* s_\ell$. Here C_ℓ transports queries backward, and C_ℓ^* transports retained score coordinates forward. This convention avoids conflating the two directions.

Theorem 1 (exact linear tangent closure). The following are equivalent:

1. A matrix M_ℓ satisfies $J_{\ell+1}^* T_\ell s = M_\ell J_\ell^* s$ for every admissible score $s \in \mathcal{H}_\ell$.
2. $U_\ell V_{\ell+1} \subseteq V_\ell$.
3. $E_\ell = 0$.

When these conditions hold, the forward matrix is necessarily $M_\ell = C_\ell^*$.

Proof. The first identity for all s is equivalent, by taking adjoints, to $U_\ell J_{\ell+1} = J_\ell M_\ell^*$. Its range condition is exactly the second assertion, and applying Q_ℓ gives the third. Conversely, $E_\ell = 0$ gives $U_\ell J_{\ell+1} = J_\ell J_\ell^* U_\ell J_{\ell+1} = J_\ell C_\ell$. Applying J_ℓ^* to the adjoint identity proves uniqueness. $\square$

Theorem 2 (optimal coefficient choice and its irreducible defect). For an arbitrary proposed forward matrix M_ℓ , define its residual query operator

$$R_\ell = U_\ell J_{\ell+1} - J_\ell M_\ell^*.$$

Then

$$R_\ell = J_\ell(C_\ell - M_\ell^*) + E_\ell, \quad R_\ell^* R_\ell = (C_\ell - M_\ell^*)^*(C_\ell - M_\ell^*) + E_\ell^* E_\ell. \quad (\text{C.1})$$

Consequently,

$$\inf_{M_\ell} \|R_\ell\|_{\text{op}} = \|E_\ell\|_{\text{op}}, \quad \inf_{M_\ell} \|R_\ell\|_{\text{HS}} = \|E_\ell\|_{\text{HS}},$$

with both infima attained by $M_\ell = C_\ell^*$.

Proof. The summands in (C.1) have orthogonal ranges, since one lies in V_ℓ and the other in $V_\ell^\perp$. The displayed Gram identity follows. It gives the lower bounds in both norms, and setting $M_\ell = C_\ell^*$ attains them. $\square$

This distinguishes a coefficient omitted by a pair-slice calculation from a genuine failure of the retained coordinates to be sufficient. For an incoming third-channel score s_3 and the unit vector e_4 selecting the outgoing fourth coordinate, the missed response is

$$\Delta r_{43} = \langle R_\ell e_4, s_3 \rangle = \langle (C_\ell - M_\ell^*) e_4, J_\ell^* s_3 \rangle + \langle E_\ell e_4, Q_\ell s_3 \rangle. \quad (\text{C.2})$$

If the probe is the canonical retained lift $s_3 = J_\ell e_3$, the second term is zero: its missed response is an error in the retained matrix, not evidence of intrinsic leakage. A specified physical third-cumulant perturbation may have an unresolved component, in which case both terms are present. The actual pair-slice formula is not presumed to be an orthogonal Galerkin projection.

In particular, late failure of a proposed two-gain rollout alone proves neither $E_\ell \neq 0$ nor that no two-coordinate closure exists.

6.3 A positive Gram defect and the number of missing observables

Because U_ℓ is an isometry,

$$C_\ell^* C_\ell + E_\ell^* E_\ell = I_{p_{\ell+1}}. \quad (\text{C.3})$$

For a merely bounded backward operator, the correct identity is instead

$$E_\ell^* E_\ell = J_{\ell+1}^* U_\ell^* U_\ell J_{\ell+1} - C_\ell^* C_\ell. \quad (\text{C.4})$$

Both follow by inserting $P_\ell + Q_\ell = I$. The positive operator $E_\ell^* E_\ell$, rather than a signed dimension count, measures failure of query closure.

There is a directly usable formulation for nonorthogonal or redundant features. Let $\Phi : \mathbb{R}^p \rightarrow \mathcal{H}_\ell$ synthesize the current features and let $\Psi : \mathbb{R}^q \rightarrow \mathcal{H}_\ell$ synthesize the pulled-back future features. Set

$$G = \Phi^* \Phi, \quad H = \Psi^* \Psi, \quad B = \Phi^* \Psi.$$

With G^+ the Moore–Penrose inverse,

$$\mathcal{D} = H - B^* G^+ B = \Psi^* (I - \Phi G^+ \Phi^*) \Psi \succeq 0. \quad (\text{C.5})$$

Indeed $\Phi G^+ \Phi^*$ is the orthogonal projection onto $\text{range } \Phi$. Under exact pushforward laws, H is the future feature Gram matrix. To compare coordinate systems, whiten the future feature metric on its nonsingular support. A raw gain response matrix expressed in nonorthogonal templates need not obey $I - M^* M \succeq 0$.

Theorem 3 (minimal local augmentation and approximation width). Keep $V_{\ell+1}$ fixed, and allow the current query space to be enlarged from V_ℓ to $V_\ell \oplus W$, with $W \subseteq V_\ell^\perp$. The minimum number of additional unrestricted linear observables giving exact one-step closure is

$$r_{\min} = \text{rank } E_\ell. \quad (\text{C.6})$$

If only r extra observables are allowed, the least possible operator-norm leakage is

$$\inf_{\substack{W \subseteq V_\ell^\perp \\ \dim W \leq r}} \|(I - P_W) E_\ell\|_{\text{op}} = \sigma_{r+1}(E_\ell), \quad (\text{C.7})$$

where singular values are nonincreasing and padded by zeros. The squared Hilbert–Schmidt optimum is $\sum_{j>r} \sigma_j(E_\ell)^2$.

Proof. Exact closure requires $\text{range } E_\ell \subseteq W$, proving the exact-rank assertion. Choosing W to span the first r left singular vectors gives the claimed upper bounds. If $r \geq \text{rank } E_\ell$, the residual is zero. Otherwise, for the operator-norm lower bound, restrict the domain to the span of the first $r+1$ right singular vectors. The rank of $P_W E_\ell$ is at most r , so a unit vector in this subspace lies in its kernel. On that vector, $\|(I - P_W) E_\ell v\| = \|E_\ell v\| \geq \sigma_{r+1}(E_\ell)$. For the Hilbert–Schmidt assertion, if u_j are left singular vectors, the captured squared norm is $\sum_j \sigma_j^2 \|P_W u_j\|^2$. The coefficients lie in $[0, 1]$ and sum to at most r , so this is at most $\sum_{j \leq r} \sigma_j^2$. The first r left singular vectors attain equality. $\square$

This is a Kolmogorov-width statement for the image of the unit query ball. If new observables must belong to a specified cheap, factored, positive, or otherwise restricted class, these unrestricted optima are lower bounds; they do not establish an efficient realization in that class. An observable given only by a fully evaluated network suffix may be mathematically low-dimensional and computationally expensive.

6.4 Why a finite projection index cannot replace the defect

In this subsection use the full-space isometry U_ℓ , not a restricted tangent operator that may have a kernel. Let $W_\ell = U_\ell V_{\ell+1}$. Then $\dim W_\ell = p_{\ell+1}$. The finite-dimensional projection map

$$P_\ell|_{W_\ell} : W_\ell \longrightarrow V_\ell$$

is Fredholm and has index

$$\text{ind}(P_\ell|_{W_\ell}) = \dim \ker(P_\ell|_{W_\ell}) - \dim \text{coker}(P_\ell|_{W_\ell}) = p_{\ell+1} - p_\ell. \quad (\text{C.8})$$

The equality is rank-nullity. It is independent of the principal angles between the two spaces. By contrast, the singular values of E_ℓ are the sines of those principal angles, including unit singular values for future directions orthogonal to the current space.

For example, in $\mathbb{R}^3$ put

$$V = \text{span}(e_1, e_2), \quad W_\theta = \text{span}(e_1, \cos \theta e_2 + \sin \theta e_3).$$

The projection index is zero for every $0 \leq \theta \leq \pi/2$, whereas the leakage norm is $\sin \theta$, ranging from zero to one. At $\theta = \pi/2$ both kernel and cokernel gain a dimension, leaving the index unchanged. Positive dimension imbalance does force at least $p_{\ell+1} - p_\ell$ missing directions; it provides no replacement for the full positive defect.

A probabilistic counterexample with an omitted fourth slice. Let $X = (X_1, \dots, X_4) \in \{-1, 1\}^4$, with

$$\Pr_\varepsilon(X = x) = 2^{-4}(1 + \varepsilon x_1 x_2 x_3 x_4), \quad |\varepsilon| < 1, \quad H_i = 1 + X_i.$$

Every proper marginal is independent of ε . Hence all means $\mathbb{E}H_i = 1$, variances 1, covariances 0, all third cumulants, all individual fourth cumulants -2 , and the retained $(2, 2)$ fourth cumulants are unchanged. The individual gain coordinates are $g_3 = 0$ and $g_4 = -2/3$ under the normalizations in the supplied note. Nevertheless,

$$\text{cum}(X_1, X_2, X_3, X_4) = \varepsilon.$$

For the positive ReLU readout $Y = \text{ReLU}(\sum_i H_i) = \sum_i H_i$,

$$\text{Var}(Y) = 4, \quad \kappa_4(Y) = -8 + 24\varepsilon, \quad \frac{\kappa_4(Y)}{3 \text{Var}(Y)^2} = -\frac{1}{6} + \frac{\varepsilon}{2}. \quad (\text{C.9})$$

To verify this, every nonempty proper Walsh moment vanishes, while $\mathbb{E}(X_1 X_2 X_3 X_4) = \varepsilon$. Expanding $(\sum_i X_i)^4$ gives $40 + 24\varepsilon$, from which $3(4)^2$ is subtracted. All laws have the same finite support and mutually equivalent positive measure classes. Thus fixed marginal/pair gains and fixed support topology can conceal a varying $(1, 1, 1, 1)$ feed. This is a counterexample for the stated class, not a claim about either reported neural network or a measurement of its r_{43} .

7 Exact finite-depth memory and minimal observable state

7.1 Block elimination without a stationarity assumption

Decompose a linear tangent state into retained and omitted components. In coordinates, allow the nonautonomous affine system

$$x_{\ell+1} = A_\ell x_\ell + B_\ell y_\ell + s_\ell, \quad y_{\ell+1} = C_\ell x_\ell + D_\ell y_\ell + t_\ell. \quad (\text{C.10})$$

For orthogonal score projections of the preceding section, the first block A_ℓ is C_ℓ^* in the preceding section's backward compression notation; the block letters in this subsection follow the conventional state-equation notation. Define

$$\Phi(\ell, k) = D_{\ell-1} \cdots D_k, \quad \Phi(k, k) = I.$$

Repeated substitution, or induction on ℓ , yields

$$y_\ell = \Phi(\ell, 0)y_0 + \sum_{k < \ell} \Phi(\ell, k+1)(C_k x_k + t_k),$$

and therefore

$$x_{\ell+1} = A_\ell x_\ell + s_\ell + B_\ell \Phi(\ell, 0) y_0 + \sum_{k < \ell} B_\ell \Phi(\ell, k+1) (C_k x_k + t_k). \quad (\text{C.11})$$

The observable memory blocks are $K_{\ell k} = B_\ell \Phi(\ell, k+1) C_k$. The unresolved initial state and unresolved forcing supply separate terms; neither is established by measuring the feedback blocks alone.

An instantaneous residual $q_\ell = B_\ell y_\ell$ is not automatically a transportable extra coordinate, since

$$q_{\ell+1} = B_{\ell+1} C_\ell x_\ell + B_{\ell+1} D_\ell y_\ell + B_{\ell+1} t_\ell.$$

For a more general register $q_\ell = M_\ell y_\ell$, first assume that the admissible joint states form a product of a retained linear space and an omitted linear space: omitted-state differences can vary independently while the retained state is held fixed. Under this hypothesis, exact affine closure with the prescribed forcing follows precisely when there exist maps $\bar{B}_\ell, \bar{D}_\ell$ such that, on the omitted-state space,

$$B_\ell = \bar{B}_\ell M_\ell, \quad M_{\ell+1} D_\ell = \bar{D}_\ell M_\ell. \quad (\text{C.12})$$

Necessity follows by considering two omitted states with the same register and independently varying their difference. Sufficiency follows by substituting

$$q_{\ell+1} = \bar{D}_\ell q_\ell + M_{\ell+1} C_\ell x_\ell + M_{\ell+1} t_\ell$$

into the first state equation.

If instead the admissible joint states lie in a constrained linear subspace $\mathcal{R}_\ell$, the correct necessary and sufficient condition is $\ker(S_\ell|_{\mathcal{R}_\ell}) \subseteq \ker(S_{\ell+1}\mathcal{T}_\ell|_{\mathcal{R}_\ell})$, where $S_\ell(x, y) = (x, M_\ell y)$ and $\mathcal{T}_\ell$ is the full linear state update. This is the factor-through-a-quotient criterion. The stronger separate factorizations in (C.12) need not be necessary on a constrained reachable graph such as $y = x$.

7.2 Schur complement and the self-energy of an omitted channel

For a stationary bounded block operator, complexifying the Hilbert spaces if necessary, $\mathcal{T} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, assume $zI - D$ is invertible and write $R_D(z) = (zI - D)^{-1}$. Solving $(zI - \mathcal{T})(x, y) = (f, g)$ first for y gives

$$[zI - A - \Sigma(z)]x = f + BR_D(z)g, \quad \Sigma(z) = BR_D(z)C. \quad (\text{C.13})$$

This is the elementary Schur-complement form of the Feshbach reduction. More explicitly,

$$\begin{pmatrix} I & BR_D(z) \\ 0 & I \end{pmatrix} (zI - \mathcal{T}) \begin{pmatrix} I & 0 \\ R_D(z)C & I \end{pmatrix} = \begin{pmatrix} zI - A - \Sigma(z) & 0 \\ 0 & zI - D \end{pmatrix}.$$

Both triangular factors are invertible. Thus, whenever the indicated inverse exists, the retained block of $(zI - \mathcal{T})^{-1}$ is $[zI - A - \Sigma(z)]^{-1}$. The same factorization identifies kernels and cokernels, so the Fredholm index of $zI - \mathcal{T}$, when defined, equals that of $zI - A - \Sigma(z)$ under the stated invertibility of $zI - D$. This index identity does not determine the entries or size of $\Sigma(z)$.

For $|z| > \|D\|$, the norm-convergent Neumann series gives

$$\Sigma(z) = \sum_{j \geq 0} z^{-j-1} B D^j C. \quad (\text{C.14})$$

The self-energy thus packages the observable memory coefficients. Its economical realization depends on which omitted directions are both reached by C and observed by B , not on the ambient size of D . The actual depth-dependent network requires (C.11); no stationarity is inferred from this resolvent illustration.

7.3 Minimal reachable–observable realization over a finite horizon

Consider the omitted subsystem as an input-output system,

$$y_{\ell+1} = D_\ell y_\ell + G_\ell u_\ell, \quad z_\ell = B_\ell y_\ell, \quad (\text{C.15})$$

where the admissible initial values and independently variable input channels are specified linear spaces. The inputs may include the retained-state feedback ports $C_\ell x_\ell$ and a declared forcing family. Write $\mathcal{R}_\ell$ for the linear subspace of reachable omitted states at depth ℓ , and define

$$\mathcal{N}_\ell = \mathcal{R}_\ell \cap \bigcap_{j=\ell}^{L-1} \ker(B_j \Phi(j, \ell)).$$

The finite observability map is

$$\mathcal{O}_\ell y = (B_j \Phi(j, \ell) y)_{j=\ell}^{L-1}.$$

At the terminal cut set $\mathcal{O}_L = 0$, $\mathcal{N}_L = \mathcal{R}_L$, and $\mathcal{Z}_L = \{0\}$.

Theorem 4 (exact minimal memory at a cut). The quotient

$$\mathcal{Z}_\ell = \mathcal{R}_\ell / \mathcal{N}_\ell$$

has well-defined time-dependent state updates and output maps reproducing all outputs for the specified input family. Its dimension is

$$\dim \mathcal{Z}_\ell = \text{rank}(\mathcal{O}_\ell|_{\mathcal{R}_\ell}). \quad (\text{C.16})$$

Let $\mathbf{R}_\ell$ map past input and initial-state coordinates onto $\mathcal{R}_\ell$, and define the past-to-future matrix $\mathbf{H}_\ell = \mathcal{O}_\ell \mathbf{R}_\ell$. Every causal linear state realization of the same input-output family has state dimension at least $\text{rank } \mathbf{H}_\ell$ at this cut; the quotient attains that bound.

Proof. If $n \in \mathcal{N}_\ell$, then $B_\ell n = 0$. Reachability is forward invariant under D_ℓ , and

$$B_j \Phi(j, \ell + 1) D_\ell n = B_j \Phi(j, \ell) n = 0 \quad (j \geq \ell + 1).$$

Thus $D_\ell \mathcal{N}_\ell \subseteq \mathcal{N}_{\ell+1}$, proving that the transition and output maps descend to the quotient. The kernel of $\mathcal{O}_\ell|_{\mathcal{R}_\ell}$ is exactly $\mathcal{N}_\ell$, which proves the dimension formula. Surjectivity of $\mathbf{R}_\ell$ onto the reachable subspace shows that this dimension equals $\text{rank } \mathbf{H}_\ell$. Any alternative r -dimensional state at the cut factors the same past-to-future map through $\mathbb{R}^r$; hence its rank is at most r . The quotient construction realizes equality. $\square$

The theorem applies to the declared input family. Treating x_ℓ as arbitrary independent inputs can enlarge reachability relative to an actual closed-loop network trajectory. Conversely, two observed trajectories do not identify the entire specified input-output map.

Individual blocks $K_{\ell k}$ of size 2×2 have rank at most two regardless of the memory dimension. The relevant rank is that of the entire cross-cut matrix $\mathbf{H}_\ell$. For example, an r -step scalar delay has one-dimensional input and output and scalar memory coefficients, but its sufficiently long-horizon Hankel matrix is an $r \times r$ anti-diagonal permutation matrix. Its rank, and hence its required linear memory dimension, is r .

For a chosen Euclidean input and output normalization, $\sigma_{r+1}(\mathbf{H}_\ell)$ is the least operator-norm error of a rank- r approximation at that cut, by the singular-vector argument in Theorem 3. It is a lower bound on the error of any r -state linear realization of those past-to-future responses. Truncating a single cut's SVD does not by itself construct a consistent causal realization at all cuts.

7.4 The exact finite-horizon query space

Let $\mathcal{Q}_\ell \subseteq \mathcal{H}_\ell$ be the query space explicitly required at layer ℓ . Define backward

$$V_L^* = \mathcal{Q}_L, \quad V_\ell^* = \mathcal{Q}_\ell + U_\ell V_{\ell+1}^*. \quad (\text{C.17})$$

Equivalently,

$$V_\ell^* = \text{span}_{j=\ell, \dots, L} U_\ell U_{\ell+1} \cdots U_{j-1} \mathcal{Q}_j.$$

This is the smallest family of linear query spaces containing all $\mathcal{Q}_\ell$ and satisfying exact backward closure $U_\ell V_{\ell+1}^* \subseteq V_\ell^*$. The proof is backward induction: any competing closed family containing the required queries must contain the right side of each recursive step.

The formal bound $\dim V_\ell^* \leq \sum_{j=\ell}^L \dim \mathcal{Q}_j$ is not an efficiency theorem. If only one final readout is required, its exact pullback is already one observable at each layer, but evaluating or integrating that observable may encode the entire expensive network suffix. An algorithmic result must also bound representation, extraction, update, and readout costs.

8 Readout-weighted error and a finite test of insufficiency

8.1 An exact adjoint error identity

For a proposed linear recursion $\hat{x}_{\ell+1} = M_\ell \hat{x}_\ell$, let the actual retained sequence satisfy

$$x_{\ell+1} = M_\ell x_\ell + \delta_\ell.$$

For a final linear readout $c^* x_L$, define

$$v_L = c, \quad v_\ell = M_\ell^* v_{\ell+1}.$$

Telescoping gives the exact identity

$$c^*(x_L - \hat{x}_L) = v_0^*(x_0 - \hat{x}_0) + \sum_{\ell=0}^{L-1} v_{\ell+1}^* \delta_\ell. \quad (\text{C.18})$$

Thus errors must be weighted by their remaining readout sensitivity; unweighted gain discrepancies need not rank candidate closures correctly.

In the tangent setting above, $\delta_\ell = R_\ell^* s_\ell$. If the initial retained coordinates match, then

$$|c^*(x_L - \hat{x}_L)| \leq \sum_{\ell=0}^{L-1} \|R_\ell v_{\ell+1}\| \|s_\ell\|. \quad (\text{C.19})$$

Under exact pushforward laws, $\|s_\ell\| \leq \|s_0\|$. More sharply, without independent score forcing, put

$$r_{\text{eff}} = \sum_{\ell=0}^{L-1} U_0 U_1 \cdots U_{\ell-1} R_\ell v_{\ell+1}, \quad U_0 \cdots U_{L-1} = I.$$

Then

$$c^*(x_L - \hat{x}_L) = \langle r_{\text{eff}}, s_0 \rangle, \quad \sup_{\|s_0\| \leq 1} |c^*(x_L - \hat{x}_L)| = \|r_{\text{eff}}\|. \quad (\text{C.20})$$

The last equality is Hilbert duality. It preserves cancellations that the sum-of-norms bound discards. Additional known forcing defects enter (C.18) separately. A nonlinear final mean or gain readout requires its specified derivative and a finite-amplitude remainder bound before this tangent estimate becomes a global error bound.

For a nonlinear proposed update F_ℓ , the elementary reference-state diagnostic remains exact:

$$r_\ell = x_{\ell+1}^{\text{truth}} - F_\ell(x_\ell^{\text{truth}}), \quad x_{\ell+1}^{\text{truth}} - \hat{x}_{\ell+1} = r_\ell + F_\ell(x_\ell^{\text{truth}}) - F_\ell(\hat{x}_\ell). \quad (\text{C.21})$$

This separates one-step closure error from autonomous state drift.

8.2 A finite probe criterion with an explicit scope

For centered admissible probe scores $s_1, \dots, s_k$, form the matrices

$$P_{ja} = \delta_{s_a} x_j, \quad R_{ia} = \delta_{s_a} x'_i.$$

There exists a linear tangent closure on this probe span if and only if

$$R = AP \text{ for some } A \iff \ker P \subseteq \ker R \iff R(I - P^+P) = 0. \quad (\text{C.22})$$

For the nontrivial direction, $\ker P \subseteq \ker R$ makes $A(Pu) = Ru$ well-defined on range P , after which A can be extended linearly. The final equivalence uses that $I - P^+P$ projects onto $\ker P$.

Every retained mean, covariance, normalizer, and context coordinate used by the proposed update must be included in P , or explicitly fixed as an external reference input. If P has full column rank, the test is vacuous. In particular, two independent gain probes can estimate a two-channel response matrix without testing whether other directions are omitted. To test a proposed extra register q , use current (x, q) and next (x', q') ; predicting only the next fourth readout does not prove that q itself transports.

Equivalently, let ϕ_j be centered current influence functions and $\psi \in \mathcal{H}_\ell$ the centered next-readout influence function pulled back to the current layer by U_ℓ . The residual score

$$s = \psi - \text{Proj}_{\text{span}\{\phi_j\}} \psi$$

satisfies

$$\delta_s x_j = 0, \quad \delta_s (\text{next readout}) = \|s\|^2.$$

A nonzero residual is a tangent obstruction when that score belongs to the declared admissible class. Bounded centered scores generate positive paths $d\mu_\varepsilon = (1 + \varepsilon s) d\mu$ for sufficiently small ε . For a general centered L^2 score, the positive path

$$\frac{d\mu_\varepsilon}{d\mu} = \frac{(1 + \varepsilon s/2)^2}{1 + \varepsilon^2 \|s\|^2/4}$$

has first derivative s , provided the needed statistic derivatives and integrability conditions hold. Passing a finite collection of probe tests proves factorization only on that collection's span and only to the tested derivative order. Estimated residual directions require uncertainty control, especially if selected from the same Monte Carlo sample used to measure them.

8.3 What remains specific to the reported networks

The general theoretical implication proved here is precise: the ordinary finite-compression index cannot determine a missing numerical fourth-order feed; the positive operator defect, its singular spectrum, and the reachable-observable continuation space are the quantities that can determine closure and its minimal linear augmentation. A scalar omitted response is one matrix element of a specified residual operator, and equation (C.2) separates coefficient correction from intrinsic omitted information.

The unresolved network-specific question is whether the relevant defect range and its future pullbacks admit a small representation in the allowed, cheaply evaluable observable class, with sufficiently small readout-weighted residual. This requires the exact retained-coordinate definitions, reference laws, source perturbations, downstream query weights, and either analytic evaluations or statistically controlled response data. Agreement of two reported trajectories does not determine those counterfactual maps, their ranks, or their uncertainty. No unseen implementation, raw output, or new chain run is used in this section.

9 An exact factor-Gaussian reduction of the missing pair-source response

The following result makes the missing one-step coefficient constructive under one explicit structural hypothesis. Suppose the Gaussian reference covariance has an exact representation

$$\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2) + BB^\top, \quad \sigma_i > 0, \quad B \in \mathbb{R}^{n \times r}, \quad 0 \leq r \leq n.$$

No low-rank approximation to Σ is being made. In particular, an exact representation may require r comparable to n , in which case the result need not give an inexpensive integration algorithm. Write the source as

$$Z_i = \mu_i + B_i X + \sigma_i \epsilon_i, \quad X \sim N(0, I_r), \quad \epsilon \sim N(0, I_n), \quad X \perp \epsilon.$$

Here B_i is the i th row of B . Conditional on X , all the Z_i are independent. This section integrates those n independent coordinates analytically, leaving only an r -dimensional Gaussian integral.

9.1 The scalar observable and its complete frozen influence

Fix one downstream row $w \in \mathbb{R}^n$. At the reference law put

$$H_i = (Z_i)_+, \quad \bar{h}_i = \mathbb{E}H_i, \quad u_i = w_i(H_i - \bar{h}_i), \quad S_k = \sum_i u_i^k.$$

Let $C_{ij} = \mathbb{E}[u_i u_j]$, let C_{off} be C with zero diagonal, and put

$$V = \mathbf{1}^\top C \mathbf{1}, \quad D_C = \text{tr } C, \quad R = \sum_{i \neq j} C_{ij}^2.$$

The letter D_C denotes a scalar covariance trace, not a Dirac operator. For the sum of the $(3, 1)$, $(2, 1, 1)$, and $(1, 1, 1, 1)$ multiplicity classes,

$$K_{\text{miss}} = \mathbb{E}P(u) - 3(V^2 - D_C^2) + 6R, \quad P(u) = S_1^4 - 3S_2^2 + 2S_4.$$

Define

$$b_i = 4\mathbb{E}S_1^3 - 12\mathbb{E}(S_2 u_i) + 8\mathbb{E}u_i^3$$

and the reference-law polynomial

$$F(u) = P(u) - b^\top u - 6V S_1^2 + 6D_C S_2 + 12u^\top C_{\text{off}} u.$$

For every differentiable probability-law tangent $\mathcal{L}$ for which these fourth moments can be differentiated, $\mathcal{L}\mathbf{1} = 0$ implies $\delta K_{\text{miss}} = \mathcal{L}F$. To see this, differentiating the centering in $\mathbb{E}P(u)$ contributes $-\sum_i \mathbb{E}(\partial_i P) \mathcal{L}u_i = -\mathcal{L}(b^\top u)$. Because $\mathbb{E}u_i = 0$, one has $\delta C_{ij} = \mathcal{L}(u_i u_j)$, hence $\delta V = \mathcal{L}S_1^2$, $\delta D_C = \mathcal{L}S_2$, and $\delta R = 2\sum_{i \neq j} C_{ij} \mathcal{L}(u_i u_j)$. Substitution gives exactly $\mathcal{L}F$. The scalar expression for K_{miss} itself follows by subtracting the (4) and $(2, 2)$ fourth-moment terms $3S_2^2 - 2S_4$ and their Wick terms $3D_C^2 + 6R$ from the full fourth cumulant $\mathbb{E}S_1^4 - 3V^2$.

In all auxiliary mean derivatives below, $\bar{h}, b, C, V, D_C$ are fixed at this reference law. Their first-variation contributions have already been incorporated into F . Differentiating them again would double-count those contributions.

9.2 Univariate half moments and their derivative jets

For $Y \sim N(a, s^2)$, $s > 0$, define

$$M_0(a, s) = \mathbb{P}(Y > 0) = \Phi(a/s), \quad M_k(a, s) = \mathbb{E}[Y^k \mathbf{1}_{Y>0}], \quad k \geq 1.$$

Writing $z = a/s$, elementary integration by parts gives

$$M_1 = a\Phi(z) + s\varphi(z), \quad M_k = aM_{k-1} + (k-1)s^2M_{k-2} \quad (k \geq 2).$$

Indeed, $yp_{a,s}(y) = ap_{a,s}(y) - s^2p'_{a,s}(y)$; integrating against y^{k-1} on $(0, \infty)$ gives the recurrence, with the boundary contribution $s\varphi(z)$ when $k = 1$.

The first three derivatives with respect to a are completely explicit:

	$\partial_a M_k$	$\partial_a^2 M_k$	$\partial_a^3 M_k$
$k = 1$	M_0	$\varphi(z)/s$	$-z\varphi(z)/s^2$
$k = 2$	$2M_1$	$2M_0$	$2\varphi(z)/s$
$k = 3$	$3M_2$	$6M_1$	$6M_0$
$k = 4$	$4M_3$	$12M_2$	$24M_1$

The relation $\partial_a M_k = kM_{k-1}$, $k \geq 1$, followed by derivatives of $\Phi(a/s)$, proves the table. It also exhibits the density and density derivative terms produced by differentiating a ReLU gate.

Introduce independent auxiliary shifts $\eta = (\eta_1, \dots, \eta_n)$ and, at a fixed outer node x , set

$$a_i(x, \eta_i) = \mu_i + B_i x + \eta_i, \quad u_i(\eta_i) = w_i((a_i(x, \eta_i) + \sigma_i \epsilon_i)_+ - \bar{h}_i),$$

$$\nu_{ik}(x; \eta_i) = \mathbb{E}_{\epsilon_i}[u_i(\eta_i)^k], \quad k = 1, 2, 3, 4.$$

Their closed formula is

$$\nu_{ik} = w_i^k \left[(-\bar{h}_i)^k + \sum_{j=1}^k \binom{k}{j} (-\bar{h}_i)^{k-j} M_j(a_i(x, \eta_i), \sigma_i) \right].$$

The constant binomial term here is $(-\bar{h}_i)^k \mathbb{E}[H_i^0] = (-\bar{h}_i)^k$, because $H_i^0 = 1$. It is *not* multiplied by the truncated quantity $M_0 = \Phi(a_i/\sigma_i)$. For positive-order derivatives that constant term has derivative zero. The moment recurrence and derivative table therefore compute $\nu_{ik}^{(q)} = \partial_{\eta_i}^q \nu_{ik}|_{\eta_i=0}$ for $1 \leq k \leq 4$ and $0 \leq q \leq 3$ with a bounded amount of work per neuron.

Equivalently, Gaussian density differentiation gives the score identity

$$\nu_{ik}^{(q)} = \sigma_i^{-q} \mathbb{E}_{\epsilon_i}[u_i(0)^k \text{He}_q(\epsilon_i)], \quad q = 0, 1, 2, 3,$$

where He_q is the probabilists' Hermite polynomial. This identity confirms that all gate-boundary contributions are included; it does not replace the closed univariate computation by an inner Monte Carlo calculation.

9.3 Seven conditional coordinates suffice for the scalar polynomial

At each (x, η_i) form the conditional cumulants

$$\begin{aligned} c_{i2} &= \nu_{i2} - \nu_{i1}^2, \\ c_{i3} &= \nu_{i3} - 3\nu_{i2}\nu_{i1} + 2\nu_{i1}^3, \\ c_{i4} &= \nu_{i4} - 4\nu_{i3}\nu_{i1} - 3\nu_{i2}^2 + 12\nu_{i2}\nu_{i1}^2 - 6\nu_{i1}^4. \end{aligned}$$

Define the seven-vector and its sum by

$$\mathbf{a}_i = (\nu_{i1}, c_{i2}, c_{i3}, c_{i4}, \nu_{i2}, \nu_{i4}, \nu_{i2}^2), \quad A = \sum_i \mathbf{a}_i, \quad m = (\nu_{11}, \dots, \nu_{n1})^\top.$$

All these objects depend on x and the auxiliary shifts. Superscripts (1), (2), (3) on $\mathbf{a}_i$ or m_i below mean derivatives with respect to the neuron-specific η_i , evaluated at zero. They can be obtained by finite polynomial algebra on the moment jets. For example, one may substitute $\sum_{q=0}^3 \nu_{ik}^{(q)} t^q / q!$ into the displayed cumulant polynomials and keep terms through t^3 ; the seven-dimensional size is fixed.

Theorem F.1 (exact conditional expectation). Define

$$P_0(A) = A_1^4 + 6A_1^2 A_2 + 3A_2^2 + 4A_1 A_3 + A_4 - 3A_5^2 - A_6 + 3A_7 \\ - 6V(A_1^2 + A_2) + 6D_C A_5.$$

Then, with the influence polynomial frozen as above,

$$G(x, \eta) := \mathbb{E}_\epsilon F(u(\eta)) = P_0(A) - b^\top m + 12m^\top C_{\text{off}} m.$$

Proof. Conditional independence makes the first four cumulants of S_1 equal to A_1, A_2, A_3, A_4 . Expanding the fourth moment in those cumulants gives

$$\mathbb{E}_\epsilon S_1^4 = A_1^4 + 6A_1^2 A_2 + 3A_2^2 + 4A_1 A_3 + A_4, \quad \mathbb{E}_\epsilon S_1^2 = A_1^2 + A_2.$$

Alternatively both identities follow by expanding a product of independent univariate moment-generating polynomials through degree four; no assumption of Gaussian conditional activations is used. Furthermore

$$\mathbb{E}_\epsilon S_2^2 = \left(\sum_i \nu_{i2} \right)^2 + \sum_i (\nu_{i4} - \nu_{i2}^2) = A_5^2 + A_6 - A_7,$$

while $\mathbb{E}_\epsilon S_4 = A_6$ and $\mathbb{E}_\epsilon S_2 = A_5$. Because C_{off} has zero diagonal, conditional independence also gives $\mathbb{E}_\epsilon(u^\top C_{\text{off}} u) = m^\top C_{\text{off}} m$. Combining these identities in F proves the formula. $\square$

For explicit implementation, let $J = \nabla P_0(A)$, $H = \nabla^2 P_0(A)$, and $L = \nabla^3 P_0(A)$. Their nonzero entries are

$$J = (4A_1^3 + 12A_1 A_2 + 4A_3 - 12V A_1, 6A_1^2 + 6A_2 - 6V, 4A_1, 1, \\ -6A_5 + 6D_C, -1, 3),$$

$$H_{11} = 12(A_1^2 + A_2 - V), \quad H_{12} = H_{21} = 12A_1, \quad H_{13} = H_{31} = 4, \quad H_{22} = 6, \quad H_{55} = -6, \\ L_{111} = 24A_1, \quad L_{112} = L_{121} = L_{211} = 12.$$

All unspecified Hessian and third-derivative entries vanish. The symbol H in this paragraph denotes a seven-dimensional Hessian only, not the activation vector.

9.4 The incoming pair-source coefficients

Let the symmetric incoming third-cumulant tensor have only the entries

$$T_{iii} = \tau_i, \quad T_{ijj} = T_{jji} = T_{jii} = \Gamma_{ij} \quad (i \neq j),$$

and zero entries when all three indices are distinct. The array Γ_{ij} is indexed by the repeated index i and the distinct index j ; it need not satisfy $\Gamma_{ij} = \Gamma_{ji}$. The associated pure Gaussian third-cumulant tangent is the linear functional

$$\mathcal{L}_T f = \left[\sum_i \frac{\tau_i}{6} \partial_{\eta_i}^3 + \sum_{i \neq j} \frac{\Gamma_{ij}}{2} \partial_{\eta_i}^2 \partial_{\eta_j} \right] \mathbb{E}_{Z \sim N(\mu + \eta, \Sigma)} f(Z) \Big|_{\eta=0}.$$

The factor $1/2$ includes the three permutations of (i, i, j) in the $1/6$ symmetric third-derivative contraction. Here η is a mean-shift probe used to evaluate the source score; the resulting third-cumulant tangent itself has zero first mean and covariance variation. Differentiating the Gaussian exponential generating function shows that its cumulant variation is exactly T in degree three and zero in other degrees. It can equivalently be realized by the positive squared-score density construction, so the tangent need not be interpreted as a finite sign-changing density.

Theorem F.2 (the factor-node formula). Evaluate all the quantities below at $(x, \eta = 0)$ and put $c = C_{\text{off}}m$. Define

$$Q_i(x) = J \cdot \mathbf{a}_i^{(3)} + 3H[\mathbf{a}_i^{(2)}, \mathbf{a}_i^{(1)}] + L[\mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}] - b_i m_i^{(3)} + 24m_i^{(3)} c_i,$$

and, for $i \neq j$,

$$Q_{ij}(x) = H[\mathbf{a}_i^{(2)}, \mathbf{a}_j^{(1)}] + L[\mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}, \mathbf{a}_j^{(1)}] + 24C_{ij}m_i^{(2)}m_j^{(1)}.$$

Then the complete missing-sector pair-source response is

$$\delta_T K_{\text{miss}} = \mathbb{E}_{X \sim \gamma_r} \left[\sum_i \frac{\tau_i}{6} Q_i(X) + \sum_{i \neq j} \frac{\Gamma_{ij}}{2} Q_{ij}(X) \right].$$

No outgoing fourth-order tensor or third-order activation table occurs.

Proof. Only $\mathbf{a}_i$ depends on η_i , and $A = \sum_i \mathbf{a}_i$. The third derivative chain rule therefore gives

$$\begin{aligned} \partial_{\eta_i}^3 P_0(A) &= J \cdot \mathbf{a}_i^{(3)} + 3H[\mathbf{a}_i^{(2)}, \mathbf{a}_i^{(1)}] + L[\mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}], \\ \partial_{\eta_i}^2 \partial_{\eta_j} P_0(A) &= H[\mathbf{a}_i^{(2)}, \mathbf{a}_j^{(1)}] + L[\mathbf{a}_i^{(1)}, \mathbf{a}_i^{(1)}, \mathbf{a}_j^{(1)}] \quad (i \neq j). \end{aligned}$$

The third diagonal derivative of $-b^\top m$ is $-b_i m_i^{(3)}$ and its mixed derivatives vanish. Symmetry and zero diagonal of C_{off} give

$$\partial_{\eta_i}^3 (12m^\top C_{\text{off}}m) = 24m_i^{(3)}(C_{\text{off}}m)_i, \quad \partial_{\eta_i}^2 \partial_{\eta_j} (12m^\top C_{\text{off}}m) = 24C_{ij}m_i^{(2)}m_j^{(1)}.$$

Thus $Q_i = \partial_{\eta_i}^3 G$ and $Q_{ij} = \partial_{\eta_i}^2 \partial_{\eta_j} G$. Theorem F.1 and conditioning express the shifted full expectation as $\mathbb{E}_X G(X, \eta)$. All derivatives through order three can be interchanged with this outer expectation: $\sigma_i > 0$ gives the displayed smooth univariate formulas, and, uniformly for η in a compact neighborhood of zero, Gaussian score identities bound the relevant derivatives by a polynomial in $|X|$. Gaussian moments dominate that polynomial. Apply the stated Gaussian tangent and the complete influence identity. $\square$

9.5 Baseline constants require only scalar, vector, and pair contractions

The exact unconditional mean is already a univariate expression:

$$\bar{h}_i = M_1(\mu_i, s_i), \quad s_i^2 = \sigma_i^2 + \|B_i\|^2.$$

The covariance diagonal is

$$C_{ii} = w_i^2 (M_2(\mu_i, s_i) - \bar{h}_i^2),$$

and for $i \neq j$ conditional independence gives

$$C_{ij} = \mathbb{E}_X [m_i(X)m_j(X)].$$

At a baseline node, the remaining contractions for b are

$$\mathbb{E}_\epsilon S_1^3 = A_1^3 + 3A_1A_2 + A_3,$$

$$\boxed{\mathbb{E}_\epsilon(S_2u_i) = \nu_{i3} + m_i(A_5 - \nu_{i2}), \quad \mathbb{E}_\epsilon u_i^3 = \nu_{i3}.$$

The first identity is the third moment-cumulant relation for a sum of independent conditional variables. In the second identity the $j = i$ term is ν_{i3} and each $j \neq i$ term factorizes as $\mathbb{E}_\epsilon(u_i u_j^2) = m_i \nu_{j2}$. Integrating these quantities over X and substituting in the definition of b_i completes the baseline construction. It never constructs $\mathbb{E}[u_i u_j u_k]$ as a three-index array.

Theorem F.3 (arithmetic and storage cost, conditional on outer integration). For a supplied exact factorization, a fixed row w , and dense pair source Γ , baseline and response integrands can each be evaluated at one outer node in

$$\boxed{O(nr + n^2) = O(n^2)}$$

arithmetic and special-function evaluations, with $O(n^2 + nr)$ storage for the supplied and accumulated matrices and $O(n)$ working storage for unary jets. No array with three or four neuron indices is required.

Proof. Computing Bx costs $O(nr)$. The four conditional moments and their first three derivatives require a bounded number of elementary operations and univariate Φ, φ evaluations per neuron. Forming the conditional cumulants, seven-vectors, jets, sums, and the fixed-size tensors J, H, L costs $O(n)$. At a baseline node the outer product $mm^\top$ costs $O(n^2)$; all contractions for b cost $O(n)$. At a response node the matrix-vector product $C_{\text{off}}m$ costs $O(n^2)$. Each Q_i costs constant arithmetic, and each Q_{ij} does also because H, L act in dimension seven and have the stated sparse entries. Accumulating the ordered pair sum costs $O(n^2)$ without storing the full array of Q_{ij} . Since $r \leq n$, the displayed bound follows. The stored covariance and source arrays have $O(n^2)$ entries, B has nr entries, and all neuron jets have bounded size per neuron. $\square$

An exact integral is not automatically evaluated exactly by finitely many quadrature nodes. If Q_0 and Q_1 nodes are used for baseline and response integration, their arithmetic cost is $O((Q_0 + Q_1)(nr + n^2))$, apart from the quadrature's own construction and precision requirements. The required node counts depend on r , accuracy, the factor loadings, and the smoothing scales σ_i ; a tensor-product rule with q nodes per factor has q^r nodes. This theorem makes no dimension-independent quadrature-error claim. Errors in the computed baseline constants must also be propagated through the response formula. For $r = 0$ there is no outer integration, and the formulas are finite univariate closed expressions with the stated $O(n^2)$ contraction cost. For multiple output rows the theorem applies per row; shared unweighted unary moments can be reused, but no $O(n^2)$ bound for arbitrarily many rows is asserted.

9.6 What this constructs, and what remains a separate theorem

Corollary F.4 (closed formula at the independent standard reference). If $Z \sim N(0, I_n)$ and the incoming source is the stated pair source, put $c = 1/\sqrt{2\pi}$. Then

$$\boxed{\Delta_T = \frac{3c}{2} \sum_{i \neq j} \Gamma_{ij} w_i w_j \left[w_i^2(1 + 4c^2) + w_j^2(1 - 4c^2) \right].}$$

All diagonal-source coefficients vanish. No integration remains in this special case.

Proof. At this reference, a diagonal Hermite source depends on one Gaussian coordinate only. Its positive squared-score probability path changes only that marginal and preserves independence.

Every mixed fourth cumulant therefore remains zero, so its missing-sector derivative is zero. A source Γ_{ij} depends only on coordinates i, j . It preserves their independence from all other coordinates. A mixed cumulant involving any of those other independent coordinates is consequently zero along that path. Only the two $(3, 1)$ components within i, j remain. Proposition 8 of Section 2 gives their derivatives as $\Gamma_{ij}(3c/8 + 3c^3/2)$ and $\Gamma_{ij}(3c/8 - 3c^3/2)$. Multiplication by their four ordered weights $4w_i^3w_j$ and $4w_iw_j^3$ gives the displayed summand. Linearity of the first variation proves the formula for their sum. $\square$

The result is an exact integral representation of the missing, matched third-cumulant pair-source feed at a specified Gaussian reference state. Under an exact small-factor covariance structure, it replaces an n -dimensional activation integral by an r -dimensional one, after analytically integrating all independent residual coordinates. Its gate terms, centering, and covariance variations are included in the proof.

To obtain a normalized gain or a moving coherent projection, apply the full derivative of that chosen coordinate to the resulting cumulant response. To obtain a depth recursion, one must additionally prove that the required reference state and factor representation are propagated accurately, or supply the observable memory that corrects their failure. ReLU does not preserve a Gaussian law merely because its input covariance has this factor form. No evidence supplied with the task establishes small factor rank for the actual conditional covariance. These are the precise remaining conditions between the constructive one-step theorem and a practical deep closure.

A single carried random-scale variable does not by itself prove that this conditional covariance has rank-one common-factor form; the residual coordinates can remain densely correlated after the scale is conditioned on.

10 The duality bridges and the remaining network theorem

10.1 A precise bridge from observable logic to dynamic closure

The Stone–Priestley programme relates predicate algebras to spaces on which those predicates are represented [6]. Its relevant question here is which distinctions between conditional states must survive a quotient. This can be made precise before choosing any topology.

Let X_ℓ be state sets, $F_\ell : X_\ell \rightarrow X_{\ell+1}$ their updates, and $\mathcal{A}_j$ the specified real-valued queries at layer j . Write $F_{j:\ell} = F_{j-1} \circ \cdots \circ F_\ell$, with $F_{\ell:\ell} = I$. Define

$$x \sim_\ell x' \iff a(F_{j:\ell}x) = a(F_{j:\ell}x') \text{ for every } j \geq \ell, a \in \mathcal{A}_j.$$

Proposition 10.1 (The coarsest future-query congruence). *The relation $\sim_\ell$ is an equivalence relation, every required current query descends to $X_\ell/\sim_\ell$, and F_ℓ induces a well-defined update on these quotients. Any other family of equivalence relations with these two properties distinguishes at least the pairs distinguished by $\sim_\ell$.*

Proof. Reflexivity, symmetry, and transitivity follow from equality of all listed query values. The $j = \ell$ condition makes current queries constant on classes. If $x \sim_\ell x'$, every future query after applying F_ℓ is one of the queries in the definition, so $F_\ell x \sim_{\ell+1} F_\ell x'$. Finally, if another quotient respects every update and every required current query, iterate its updates and then apply the query at layer j . Related states must agree on every expression in the definition of $\sim_\ell$. This proves the claimed coarseness. $\square$

For an explicit Boolean predicate realization, let $\mathcal{B}_\ell$ be generated by the sets

$$\{x : a(F_{j:\ell}x) \leq q\}, \quad j \geq \ell, \quad a \in \mathcal{A}_j, \quad q \in \mathbb{Q}.$$

Equality on these predicates is exactly $\sim_\ell$, because rational thresholds separate real numbers. Moreover $F_\ell^{-1}\mathcal{B}_{\ell+1} \subseteq \mathcal{B}_\ell$. Thus the logical quotient and dynamic closure refer to the same future distinctions. In a finite state space the corresponding Boolean spectrum is the set of nonempty atoms. In an infinite state space its Stone spectrum may contain ultrafilters not represented by an original physical state; an additional representation argument is needed to identify the whole spectrum with a physical quotient.

The linear tangent version of this proposition is the backward query closure of Section 6. Its reachable–observable quotient identifies the exact omitted state differences that no future readout can distinguish. That is the rigorous bridge to the duality viewpoint.

Skew Boolean section algebras give a useful formal model for retaining a local value together with the domain on which it is available [7]. For a finite discrete gate base, a partial section assigns a value to each point in its domain; ordered restriction and override distinguish its value from its domain. Passing to domain data forgets the chosen local section. This explains why remembering a gate cylinder need not retain the conditional numeric state. A concrete application must still identify the fibers, restriction maps, and updates. The name of the algebra does not supply the weights in a cumulant response.

Similarly, a Heyting or bitopological interpretation requires an actual predicate algebra and its extension order. Declaring one topology to be “past” and another “future” does not prove the required duality hypotheses. The future-query congruence above specifies exactly the predicates a successful logical model must preserve.

10.2 The noncommutative object that measures the lost response

Three different operator constructions have been established, and they answer different questions.

First, the Gaussian Dirac construction supplies a spectral resolution of source tangents and readout influences. The relevant coefficient is

$$\Delta_T = \langle S_T, \Pi_3 \Psi_{\text{miss}} \rangle.$$

The finite-energy projector Π_3 and source–query Gram matrix retain the amplitude. An ordinary index records a stable signed dimension and loses this information. The gate regularity result additionally shows why local smooth-index formulas cannot be imported without handling the gate boundaries.

Second, compression of a propagation operator gives the positive multiplication defect

$$PA^*(I - P)AP = (QAP)^*(QAP).$$

This is a continuous, quantitative obstruction. The same discarded intermediate propagation appears as a memory term when omitted coordinates are eliminated. In a stationary bounded-block model it contributes the Feshbach–Schur self-energy

$$\Sigma(z) = B(z - D)^{-1}C.$$

The block factorization proving this identity is included above; the broader operator-theory context is [4]. The finite-depth memory derivation is equally compatible with the neural-network Mori–Zwanzig formulation [5], without assuming its contraction hypotheses for this deterministic network.

Third, a changing retained projection has a Grassmann connection and curvature. This correctly tracks moving templates and conditional reference frames. A globally specified independent feature list trivializes its finite-dimensional bundle. Curvature can still be nonzero, but that fact alone does not produce a missing topological class or the required amplitude.

These constructions give an explicit response operator, an explicit loss operator, and an explicit coordinate derivative. They make the NCG bridge operational while keeping the particular role of the index precise.

10.3 What the next theorem for these networks must establish

Let $T_\ell^{(3)}$ denote the exact unit incoming third-gain tangent used in the existing ledger, a_ℓ the outgoing coherent projection, and μ_ℓ, Σ_ℓ its reference parameters. The one-step correction to evaluate is

$$\Delta r_{43,\ell} = a_\ell^\top (\delta_{T_\ell^{(3)}} K_{\text{miss},k})_k,$$

with the derivative of a_ℓ added when it moves. This is now a specified analytic functional of the reference, source, and weights. It has the exact score, gate-facet, and conditional-factor representations proved in this note.

A theorem about the *deep profile* additionally needs one of the following kinds of control:

1. Exact closure: the required backward influence queries remain in a specified small retained space for every layer and every allowed perturbation.
2. Controlled approximate closure: singular values of the actual projection defects and the future-observable memory are small in norms calibrated to the final readout, with accumulated remainders bounded.
3. A structural conditional model: an exact or quantitatively controlled low-dimensional conditional representation permits the missing response and its transported queries to be evaluated with a proven error bound.

The conditional-factor theorem supplies a constructive instance of the third possibility under its explicit covariance hypothesis. It does not establish that the actual conditional state has small factor rank. A fitted 2×2 response matrix and two matching network trajectories do not imply any of these universal properties.

A useful exact consistency relation. For the full regular Gaussian scale direction, the sum of the mean/covariance drift and both cumulant channels must equal the Ward derivative in Section 4. This gives a check on a completed transport ledger that is stricter than setting row sums to one. A pair-restricted incoming source must be distinguished from that full scale direction when making the comparison.

Why an improved gain coefficient need not improve the final mean. The final-readout adjoint identity in Section 6 weights each local defect by its actual future sensitivity, including cancellations. A fitted coefficient can partly compensate for omitted sectors, moving templates, or reference-state error. Replacing it by a more accurate isolated physical derivative need not improve the final prediction until those other terms are accounted for. The note’s operator identities describe this issue without asserting that any new chain run has been made.

10.4 Evidence, review, and scope of the closeout

The source material for the project-specific setting consists of the user-supplied Note XIX and accompanying pasted notes. The network weights, the precise source-tangent implementation, gain-projection arrays, and raw Monte Carlo state data were not supplied. Consequently this is a closeout of the present *general theory investigation*, not a numerical closeout of the two networks.

The proof development was checked in separate routes. An independent reviewer reconstructed the Ward cumulants and density scores, the closure and minimal-memory proofs, the gate-facet coefficients, the explicit saturation constants, and all four results of the conditional-factor route. Corrections made during review include the independence assumption for random scaling, the product-state hypothesis for a simple register factorization, the loss of isometry for restricted tangent spaces, the terminal convention in finite-horizon observability, and the distinction between a current influence and its pulled-back future counterpart. A separate functional-analytic review checked the Gaussian spectral triple and fractional gate regularity; the bounded-transform proof was sharpened to use first-order resolvents and avoid an unstated second-order domain assumption.

The executable exact checks cover the bulk polynomial identity in dimensions one through four and two independent analytic ReLU witnesses, using set partitions and rational polynomial arithmetic. They verify those bounded calculations; they do not replace the general proofs, authenticate external theorems by themselves, or establish empirical network performance. The source archive contains the script, recorded results, provenance manifest, and review notes.

No claim of novelty in the literature is made. The external theorem checks were confined to the listed primary sources, their stated hypotheses, and the translations needed here. The explicit cumulant, gate, scale, and finite-horizon formulas are proved in the note.

The unresolved network-specific statement is now sharp: determine the missing source-readout contractions in the actual coordinate convention, and establish whether their backward observable span or memory admits the required inexpensive approximation. Its governing objects are the proved response and defect operators. An index can be useful if a new comparison theorem links it to those objects, but such a comparison has not been established merely by constructing the index-one Gaussian cycle.

References

- [1] Alain Connes, *Noncommutative Geometry*, Academic Press, 1994. Author-hosted text. Checked I.3 and IV.8, especially Proposition 18 and Theorems 19, 21–22. <https://alainconnes.org/wp-content/uploads/book94bigpdf.pdf>.
- [2] Ezra Getzler and Andras Szenes, *On the Chern character of a theta-summable Fredholm module*. Author-hosted manuscript, defining hypotheses and Theorems A–D. <https://bpb-us-e1.wpmucdn.com/sites.northwestern.edu/dist/c/2278/files/2019/08/char.pdf>.
- [3] Alain Connes and Henri Moscovici, *The local index formula in noncommutative geometry*, 1995. Author-hosted text. https://alainconnes.org/wp-content/uploads/localtrace95.ps_.pdf.
- [4] Genevieve Dusson, Israel Michael Sigal, and Benjamin Stamm, *The Feshbach–Schur map and perturbation theory*, arXiv:2105.02058. <https://arxiv.org/abs/2105.02058>.
- [5] Daniele Venturi and Xiantao Li, *The Mori–Zwanzig formulation of deep learning*, arXiv:2209.05544v4, 2023. <https://arxiv.org/html/2209.05544v4>.
- [6] Mai Gehrke and Sam van Gool, *Topological Duality for Distributive Lattices: Theory and Applications*, arXiv:2203.03286v5, 2023. <https://arxiv.org/abs/2203.03286v5>.
- [7] Andrej Bauer and Karin Cvetko-Vah, *Stone Duality for Skew Boolean Algebras with Intersections*, arXiv:1106.0425. User-attached source copy. <https://arxiv.org/abs/1106.0425>.